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HAIR COLORING VARIEGATION DEVICE AND METHOD OF USE

Abstract

A device for selectively entraining hair strands from the scalp has at least one hooking or a clamping applicator, the at least one hooking applicator employing a hook that rotates to entrain the hair strands, and the clamping application employing a flat plate. A hair product container having a styling product or hair color therein is provided. The device also includes a mechanism to apply the hair product to the entrained hair stands. The mechanism using a hooking applicator can include a worm gear.

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Background/Summary

FIELD OF THE INVENTION

[0001] The present invention relates generally to the coloring of human hair, and more particularly, to an improved method and device for quickly and effectively coloring human hair in a variegated manner.

BACKGROUND ART

[0002] Hair color variegation is a popular service performed by the professional beauty industry. The process involves the segregation of one or more sections of human hair followed by the treatment of the segregated hair with a hair coloring method or chemical. The technical skill required to separate particular sections of a person's hair from the remainder has kept this procedure mostly in the purview of hair salons.

[0003] A previously popular method for highlighting hair is described in U.S. Pat. No. 5,562,111. The method disclosed therein involves a cap tightly fitted over a scalp of combed-back hair. Strands of hair are then pulled through holes in the cap with a crochet hook and the exposed hair is colored to create the effect of variegation. Although this method can be somewhat successful both at keeping the chemical hair coloring from bleeding onto the hair not intended for treatment and creating a generally variegated look, the necessity of drawing hairs through individual holes in the cap makes it difficult for the technician to consistently draw out a section of hair from the desired area without unintentionally entraining undesired sections of hair from areas surrounding the hole. The end result is unpredictable and, sometimes, very undesirable. Moreover, the available variegation pattern is dictated by the location and distribution of the holes in the cap. Additional disadvantages to this method include the inability to effectively color hair roots, the inability to consistently prevent the bleeding of color to adjacent sections of unselected hair, and the pain experienced by the recipient due to the repeated pulling of her hair through small holes. U.S. Pat. No. 4,165,754 is another example of a hair highlighting method employing a cap over the scalp. This method has the identical drawbacks of the '111 patent.

[0004] Alternatively, there are various combing methods used to apply hair color in a variegated manner. A general method involves dipping a comb into a liquid hair color and pulling the comb through the hair to be treated. Only relatively large sections of hair can be treated in this manner and it is difficult for the operator to avoid color bleeding onto hair not intended for treatment. U.S. Pat. No. 3,349,781 describes a method wherein a hair stylist parts hair into sections and uses a brush with a series of spaced tufts to brush streaks onto random strands. The tufts of the brush are dipped into a hair color composition and retain the composition until the brush is drawn across the strands to be colored, thus depositing the colorant thereon. This method utilizes protective sheets placed under and over the streak-treated partings before and after treatment to avoid color bleeding to adjacent hair. However, using this brush method makes it difficult to choose which strands of hair will be treated. Hence, there is minimal control over the placement of the hair treatment. Therefore, larger sections of hair are treated, resulting in a more unnatural hair coloring effect.

[0005] U.S. Pat. No. 5,337,765 describes a modular brush for applying hair color compositions with a brush body and detachable bristle modules so that the brush can be configured to achieve a user-defined variegated pattern. However, this arrangement presents the same limitations as described above for the '781 patent.

[0006] A more commonly used technique by those skilled in the art involves selecting hair through weaving with a conventional tail comb and then placing the selected sections onto aluminum foil (or some other sheet of barrier material) and then painting sections with a hair color composition. A dispensing device for metallic foil that may be used in this process is disclosed in U.S. Pat. No. 6,237,608. The foil method allows for smaller, more independent, more consistently variegated sections to be treated closer to the scalp, resulting in a more naturally variegated final appearance. When using this method, the potential for color bleeding onto surrounding hair is reduced. But even with these advantages over other hair color variegation techniques, the foil method is very time consuming and expensive. For an average client, approximately 30 to 50 minutes is required to complete this method of hair coloration.

[0007] Hair color variegation techniques that involve color treated sections that have been woven away and placed inside a barrier material for processing produce natural and attractive variegated appearance. It follows then that advancement in the field of hair color variegation involves weaving, color treatment and barrier material. Reference will now be made to technology that attempts to advance on one or more of these three general systematic elements.

[0008] U.S. Patent Application No. 2005/0028835 discloses “A Device for Dispensing a Barrier Material to a Lock of Hair.” This device (although some of the embodiments vary greatly) is comprised of two tape dispensers that are hinged at the roll end. The tape dispenser end (distal to the roll end) opens and closes in such a way as to cause the faces of the two tapes to touch. A section of hair can be chosen and encapsulated between the two tapes. The face of one or both of the tapes is treated with one or both of the chemical hair color components. The embodiments also include means within the device to apply hair color just before the hair is encapsulated within the tape. This method, although saving time and product, still lacks the ability to automatically, quickly and accurately weave away a plurality of selected hair sections for variegation purposes.

[0009] U.S. Pat. No. 5,152,306 discloses a hair-weaving comb that has regular teeth and inwardly barbed teeth attached alternately across the spine of the comb. In practice, a thin section of hair is parted away from the scalp. The teeth of the comb are then pushed into the parting and drawn back out. The barbed teeth pick up sections of hair while the straight teeth do not. An operator grabs the hooked hair, pulls the comb away and lets the non-hooked hair fall. This device allows for a faster and more consistent weave than the manual hair weaving method. However, it does not offer any device or method to apply color or barrier material. In addition, the device does not effectively pick up sections of hair in a predictable manner, nor does it pick up hair against a curved scalp surface.

[0010] U.S. Pat. No. 5,024,243 discloses a comb/color applicator combination. The device discloses a comb with a hollow spine that screws onto a container filled with chemical color composition. When the container is squeezed, the chemical composition fills the hollow spine of the comb and exits the spine through small holes positioned in between the teeth of the comb. Although this device will yield a variegated hair color appearance, there is a substantial risk of color bleeding because the variegated hair is not woven away from the rest, and the device fails to provide the technician with a high degree of control or accuracy.

[0011] U.S. Pat. No. 5,303,722 describes a hair lightening method involving the use of an optical photosensitizer and a compound capable of providing a hydrogen radical (ethanol is preferred) in a solution. The solution is applied to the hair and then left to saturate for 5 to 60 minutes. Low intensity ultraviolet light (typically provided by a comb or hood) is then applied to the hair causing a hydrogen to be exchanged between the two components in the solution, thereby creating hydrogen peroxide inside the hair shaft. The peroxide is excited by the light causing some of the hair pigment (melanin) to be destroyed. As a result, the hair subjected to the process is lightened. Using this same photochemical reaction, the '722 patent describes a method whereby the entire head of hair is saturated with the photosensitive solution followed by the segregation of small sections of hair by manual weaving. The non-segregated hair is masked with an opaque material so that only the segregated hair is exposed to the low intensity ultraviolet light. The result is “highlight” effect among the segregated hair strands. The techniques described in the '722 patent involve considerable time and manual labor.

[0012] U.S. Pat. No. 4,325,393 discloses a hooking mechanism for hair coloration. The implement has a plurality of equidistantly spaced, accurate hook members movable between open and closed positions with respect to the bottom surface of the body of the implement by an operating slide member at its top. After thus hooking and engaging spaced groups of hair strands for treatment, the implement is lifted from the scalp to isolate the strand groups for bleach or dye treatment. This implement does not offer the operator nearly the degree of control that is inherent in the instant invention. Although the bottom surface of the device is curved, it does not flexibly conform to the curve of the head. This prohibits the device from uniformly selecting portions of hair.

[0013] Furthermore, the '393 patent offers no means by which the hooked hair can have a comfortable tension applied to it when the hooks are in the closed position. Hair may be hooked away from the scalp, but it cannot be held against tension—it will simply slide through hooks when the operator pulls the device away from the head. Finally, the '393 patent does not include any means by which it can apply color compositions nor any means to assure a safe and controlled contact with the scalp by the swinging hooks.

[0014] U.S. Patent Application No. 2006/0042643 discloses a hair highlighting tool. However, the disclosed invention does not address the multiple problems overcome with the instant invention. In fact, it may exacerbate some of the problems regarding the regulation and control of hair coloration.

[0015] U.S. Pat. No. 7,530,358 overcomes many of the problems identified above but does not address the problem of applying color of higher viscosity. The ‘358’ patent does not provide means for expelling high viscosity liquid hair color from a color container onto entrained sections of hair in a controlled manner as does the present invention. The ‘358’ patent discloses a hook that is only useful for entraining hair against an applicator that distributes low viscosity hair color onto the entrained section by way of a ‘wicking’ action. The present invention features a hook and applicator arrangement that, when in the closed position, channels high viscosity liquid hair color onto entrained sections in a controlled manner. Also, the ‘358’ hooking mechanism is prohibitively complicated and relies on a mechanism that raises the hook and entrained section of hair up to the applicator. The present invention eliminates the need for this mechanism without losing function. Furthermore, the ‘358’ patent describes a mechanical means responsible for confining the hooks to a light controlled contact with the scalp. This mechanical means consists two feet separated into four scalp contact points; two contact points in front of the hook and two contact points in back of the hook creating a hook channel that extends flush with the rotation of the hook toward the scalp. The present invention involves an arrangement that likewise confines the hooks to a light, controlled contact with the scalp also employing two ‘feet’ with two scalp contact points positioned in front of as well as in back of the hook. The present preferred embodiment of the device entrains and gathers the entrained section of hair differently employing a ‘scissor action’ by gathering the hair as it approaches the closed position between the inside of the hook and the side edges of the scalp contact points or ‘feet’. Considering there are feet that only occupy the width of the applicator nozzle, this leaves the entire pivot of the hook toward the nozzle in full view of the operator. This more open hook arrangement allows the operator a better view of the entraining of the hair as well as a better view of the application of color onto the entrained section than is allowed in the ‘358’ patent. Finally, unlike the ‘358’ patent, the present invention keeps the color components separated as it dispenses and mixes them just before the color comes into contact with the entrained sections.

[0016] All of the above-cited prior art addresses certain needs. However, none solves the time, consistency and control problems that are encountered when performing the manual hair color variegation technique presently most popular in the purview of the hair salon. In addition, none have successfully combined mechanical elements into a single device to give it the ability to do all that is mentioned in the present disclosure. Accordingly, there is a need for a hair coloration device that safely, accurately, predictably, and quickly applies low and high viscosity colorant to uniformly selected and entrained portions of hair.

SUMMARY OF THE INVENTION

[0017] The present embodiment of the hair color variegation device features a pre-loaded color container that slides into the front of the handle as well as hair entraining and color dispensing mechanisms that are engaged in sequence by a single squeeze of the handle. The device is used in one hand by drawing a parting of hair across the scalp with a rod-like member extending away from the rear of the device. This member is called the parting stem. The device is then turned so that the head of the device is facing and in line with the parting of hair. The head of the device is placed along the parting so that the parting is visible 1/16″ to ¼″ or farther above the line of the contact points of the head of the device. The head of the device is now urged against the parting, at which point the head of the device will conform to the curve of the scalp. In this conformed placement, each hook is now in the correct position to accurately lift hair against each applicator nozzle. While holding the device lightly conformed to the scalp, the operator slowly squeezes the handle. As the operator slowly squeezes the handle, the hooks pivot simultaneously across the scalp, painlessly entraining sections of hair against the applicator nozzles. As the operator continues to slowly squeeze the handle, the hooks remain engaged while the squeeze plate begins to put pressure on the color container. This allows the operator to hold and slide the entrained sections of hair without applying the hair color. Continuing to squeeze the handle, the pressure of the squeeze plate onto the color container causes the liquid color to move out of the color container and therefore out of the applicator nozzle and onto the entrained hair. Now, the operator may carefully pull the device away from the scalp while maintaining a controlled pressure on the handle. In this manner, hair color is evenly deposited onto the entrained sections of hair. At this point the operator may stop applying squeeze pressure while continuing to holding the entrained sections of hair. While the hair is still entrained in one hand and no color is being deposited, the operator may place barrier material over the color treated sections with the free hand or simply let the color treated hair drop back into the rest of the hair. This application process may be repeated many times in one variegated hair color service.

[0018] Variations of the device may employ a single hooking applicator as well as any number of hooking applicators up to six or more hooking applicators. Certain variations of the device that employ one and perhaps up to three hooking applicators will not need a curvature conformation feature.

[0019] Hooking applicators vary in size allowing embodiments of the device to entrain individual sections of hair of varying size.

[0020] Other embodiments of the device feature a variety of detachable head units. This allows a single device handle to accommodate a variety of head units each featuring different numbers and sizes of hooking applicators.

[0021] Still other embodiments allow the operator to restrict the flow of hair color to some of the hooking applicators while allowing color to flow through others while the device is in use, while other embodiments provide a mechanical alternative to the rack and pinion gear drive that pivots the hook/hooks in the form of a lever system.

[0022] In earlier embodiments of the invention, each hook pivots from an open position distanced away from each corresponding applicator to a closed position that finds each hollow of each hook closed around the bottom of each applicator. As the hook moves from the open to the closed position, a section of hair has become entrained into the closed position between the hook and applicator, the entrained section of hair becomes automatically positioned directly under the exit opening of the applicator. Now, as the color exits the opening, the color is deposited directly onto the top of the entrained section of hair. As the operator draws the device away from the scalp, the entrained sections remain entrained and therefore convey through the closed position. Hair color is thereby deposited in the form of a continuous bead of color onto each section.

[0023] Additional embodiments of the invention include purpose of these additional features to

saturate the deposited bead of hair color completely into the entrained sections prior to exiting the closed position. These additional structural features close around the entrained sections of hair and, in this closed position, are referred to as a saturation chamber of the device. This saturation chamber constitutes the second stage of color application onto the entrained sections of hair and is in addition to the first stage of color application occurs in the initial ‘closed position’ of the device.
[0024] In addition to the saturation chamber embodiment, a ‘comb’ type embodiment of the device will be presented as well as an embodiment that includes mechanical features that allow the operator to adjust the size of the hair section that the hooking applicator entrains.
[0025] Other embodiments includes: a modified saturation chamber that uses seals to create a space to hold more color for hair treatment; a saturation chamber with a modified foot bridge configuration to allow more travel of the hook to reduce hair pinching; a one piece top handle and head mount for improved rack movement; and improved rack slide mechanism; a mechanism to adjust the distance of travel of the hooks via the rack slide assembly; a hinged head hood to cover the head mechanism for safety, aesthetics, and ergonomics; and a manifold retractor for moving the manifold out of the way for installing and/or removing a color container.
[0026] Another embodiment includes a hooking applicator that uses a worm gear for rotating the hooks.
[0027] Yet another embodiment includes an applicator that uses a clamping action rather than a hooking applicator for hair color application. This applicator includes opposing plates that clamp on a section of entrained hair for coloring. The plates can be sized to color a single wide swath or section of hair as opposed to multiple and spaced apart sections with an applicator using a number of hooks. This embodiment can also employ a seal and squeegee like the hooking applicator using one or more hooks. The clamping applicator can also be used by an individual for coloring the individual's hair rather than another person coloring the individual's hair.
[0028] Still another embodiment of the invention includes a modified hair color container that has dry bleach and a liquid hair color pouch, the pouch rupturable for mixing with the dry bleach. A second pouch embodiment is a dual component pouch, one component pouch holding the dry bleach and the other component pouch having the liquid. The pouches are connected together by a neck, which can be clipped closed and then opened by removing the clip so that the dry bleach and liquid can mix together.
[0029] Another feature of the invention is the ability to use hair products like styling products with the device rather than a hair coloring.
[0030] Other features of the invention include the use of releasably attachable sections for the device. In this embodiment, one section is configured to receive the hair product container and the other section houses the mechanism that allows the hair product to be conveyed to the hooking or clamping applicator.
[0031] The devices can also be used to apply a coating to one or more elongated flexible members. The coating can be a reactive coating using one or two chemically reactive components that would occupy the hair product container, either in a single or dual pouch mode. The elongated flexible members could be synthetic fibers for wigs or other hairpiece, wire, cord, animal hair, rope, cable, and the like.
[0032] Another embodiment provides an improved comb design, wherein the comb includes a squeegee and hook insert to improve application of a hair product to hair or other material.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0033] The following Brief Description of Drawings as well as the Detailed Description of Drawings that follows repeatedly refer to the following terms: open position and closed position.

Open position refers to the hook **2***a* as it appears having pivoted away from the applicator nozzle **11***a* (see FIG. **1**A). The closed position refers to the hook **2***a* as it appears having pivoted into contact with the applicator nozzle **11***a* (see FIG. **1**B).

[0034] FIG. **1**A is a front perspective view of a single hooking applicator **1***a* of the preferred embodiment. This figure depicts the hooking applicator as it appears with the hook **2***a* in the open position.

[0035] FIG. **1**B is a front perspective view of a single hooking applicator of the preferred embodiment. This figure depicts the hooking applicator as it appears with the hook **2***a* in the closed position.

[0036] FIG. **1**C is a front view of the front of an applicator nozzle **11***a* and a foot **4***a*. This figure depicts the preferred location of the scissors edge **4***f* of the foot **4***a*.

[0037] FIG. **1**D is a front view of a hooking applicator **1***a* and depicts a relocation of the scissors edge **4***f* of the foot **4***a*.

[0038] FIG. **2**A-**2**C are bottom views of a single hooking applicator of the preferred embodiment, each depicting the hook point in the closed position with the hook point positioned in the front, back and middle of the hook respectively.

[0039] FIG. **3**A is a magnified front perspective view of the preferred embodiment depicting the hook **2***a*, nozzle **11***a* and feet **4***a* in the open position and provides a detailed depiction of the geometry of each.

[0040] FIG. **3**B is a magnified front perspective view of the preferred embodiment depicting the hook **2***a*, nozzle **11***a* and feet **4***a* in the closed position with a section of hair entrained therein.

[0041] FIG. **3**C is a magnified front perspective view of the preferred embodiment depicting the hook **2***a*, nozzle **11***a* and feet **4***a* in the closed position and provides a detailed sectional depiction of the geometry of each.

[0042] FIG. **3**D is a magnified front perspective view of the preferred embodiment illustrating the hook **2***a*, nozzle **11***a* and feet **4***a* in the closed position and shows an alternative embodiment of the feet **4***a*.

[0043] FIGS. **4**A-**4**D are magnified bottom views of the preferred embodiment of hooking applicator together providing a serial depiction of the mechanical process of hair section entrainment.

[0044] FIG. **5**A is a magnified front perspective view of an alternative embodiment of the hook **2***a*, nozzle **11***a* and feet **4***a* in the open position and depicts an alternative arrangement of the hair channel **2***b* and color channel **2***c*.

[0045] FIG. **5***b* is a magnified front perspective view of an alternative embodiment of the hook **2***a*, nozzle **11***a* and feet **4***a* in the closed position. This figure depicts the alternative arrangement of the hair channel **2***b* and color channel **2***c* shown in FIG. **5**A with a section of hair entrained within.

[0046] FIGS. **6**A-**6**B are magnified front perspective views of an alternative embodiment of the hook **2***a*, nozzle **11***a* and feet **4***a* in the open and closed positions respectively. These figures illustrate yet another alternative arrangement of the hair channel **2***b* and color channel **2***c*.

[0047] FIGS. **7**A-**7**B are magnified front views of an alternative embodiment of the hook **2***a*, nozzle **11***a* and feet **4***a* in the open and closed positions respectively. These figures depict an alternative arrangement of the hook **2***a* and nozzle **11***a* featuring a hook tooth **3***b* and nozzle seal **11***b*.

[0048] FIG. **8** is a front perspective view of the complete mechanical arrangement of the preferred embodiment. In order to provide a clearer depiction, this figure includes isolated duplicate views of three components arranged around the complete view.

[0049] FIGS. **9**A-**9**D are front views of the preferred embodiment depicting, respectively, a sequence of functional interaction of said embodiment with a parting of hair **1***c*. Concurrently, these figures depict the positions and relationships of the individual mechanisms in each of the four depicted stages of a single cycle of mechanical engagement.

[0050] FIGS. **10**A-**10**D are side views of a complete assembly of the preferred embodiment depicting a sequence of functional interaction of said embodiment with a parting of hair **1***c*. Concurrently, these figures illustrate the positions and relationships of the individual mechanisms in each of the four depicted stages. These depicted stages combine to illustrate a single cycle of mechanical engagement including entrainment of hair sections and dispensing of hair color respectively. In order to demonstrate scale as well as how said embodiment may be held while in use, a hand is depicted holding said embodiment in a functional manner.
[0051] FIGS. **11**A-**11**B are perspective side views of alternative embodiments of the color container and manifold hose system featured in the preferred embodiment depicted in FIG. **8**.
[0052] FIG. **12**A is a front perspective view of a complete assembly of the preferred embodiment depicted in FIG. **8** including additional mechanical functions located at the front of the device. One additional mechanism depicted allows the operator to interrupt the flow of liquid color to one or more hoses along the manifold while allowing flow to others. The second of said functions allows the slide actuator tine/tines **93** to slide telescopically within the slide tine seat **94**.
[0053] FIG. **12**B-**12**C are side views of the additional mechanical functions depicted in FIG. **12**A. These views illustrate the relative positions of the individual mechanisms involved in each of said additional mechanical functions as the mechanisms appear in the disengaged and engaged positions respectively.
[0054] FIG. **13** depicts an alternative mechanical assembly of the head **14***a* of the preferred embodiment illustrated in FIG. **8**. This assembly utilizes an angled position of each hooking applicator **1***a* along the head **14***a* of said preferred embodiment in order for the pivoting action of each hook **2***a* to avoid being interrupted by the applicator nozzle **11***a* of each adjacent hooking applicator **1***a*.
[0055] FIGS. **14**A-**14**B depict an alternative mechanical arrangement of the hooking applicator **1***a* depicted in FIG. **1**A-**1**B. This hooking applicator **70** embodiment utilizes a hook slide **61** and hook lever **65** arrangement as a mechanical means to pivot the hook **2***a*.
[0056] FIGS. **15**A-**15**B illustrates a complete embodiment of the device while providing an alternative assembly to the preferred embodiment depicted in FIG. **8**. This alternative embodiment possesses an identical functional action as the FIG. **8** embodiment while utilizing an alternative mechanical means that integrates a device head **14***a* comprised of a row of lever action hooking applicators **70** described in FIGS. **14**A-**14**B.
[0057] FIGS. **16**A-**16**B depict front views of the lever action hooking applicator embodiment depicted in FIGS. **15**A-**15**B. The present figures provide a view of the action of the individual mechanisms located at the front of the device as they function in sequence to pivot the hooks.
[0058] FIG. **17** is a top view of the device illustrated in FIGS. **15**A-**15**B. This view provides a more complete description of the mechanical function described in FIGS. **14**, **15** and **16**.
[0059] FIG. **18** depicts an alternative mechanical arrangement of the hooking applicator described in FIGS. **1**A-**1**B. This hooking applicator illustration describes the mechanical means necessary to render an embodiment of the device capable of dispensing a stick or sticks of hair color chalk, mascara, etc. onto an entrained section/sections of hair.
[0060] FIGS. **19**A-**19**E depict five of the many possible head arrangements that may be assembled in order to give the operator various options for the final hair color variegation appearance. These figures combine various hook/hooking applicator sizes with varying distances between the hooks/hooking applicators to give the operator the opportunity to provide the hair color variegation service recipient with choices pertaining to the size of the sections being treated (by varying the size of the hook) and the distance between the treated sections (by varying the distance between the treated sections).
[0061] In addition to the option of having multiple hair color variegation devices, each with a fixed head of a different hooking applicator orientation, the operator may also be given the opportunity to have one of said device body along with several different detachable heads. FIGS. **20**A-**20**B as well

as FIGS. **21**A-**21**C and FIGS. **22**A-**22**F depict mechanical assemblies of varied sophistication, thereby, providing a range of opportunity for embodiments with such a feature.
[0062] FIG. **20**A-**20**B depicts a head and manifold of said device that detaches by disengaging the roller couplings from the head mounts as well as the rack gear from the rack slide. FIG. **20**A shows a side view of this arrangement in the attached position while FIG. **20**B depicts a side view of the detached position.
[0063] FIG. **21**A-**21**C depicts a head, manifold, head mount and rack slide assembly that detaches from the top hinge section of said device. In addition to the detachable head, this figure shows actuator tines removable from the bottom hinge section. FIG. **21**A shows a side view this arrangement in the attached position while FIG. **21**B depicts a side view of said device detached position. FIG. **21**C depicts a side view of the mechanical means of removability.
[0064] FIG. **22**A-**22**E illustrate top views of said device including the mechanical assemblies that allow the head mounts, rack slide and actuator tine assemblies to be adjustable rather than removable while the head and manifold remain removable. FIG. **22**F illustrates a bottom view of said device illustrating an adjustable actuator tine assembly.
[0065] FIGS. **23**A-**23**C and **24**A-**24**D depict a preferred embodiment of the hair color variegation device in the form of a compact, single hooking applicator, pen or marker like appliance. FIG. **23**A depicts a perspective side view of a complete assembly of said device. FIGS. **23**B and **23**C are close up front perspective views depicted to describe the mechanical assemblies involved in the pivoting of the hook from the open to closed position respectively. FIGS. **24**A-**24**D represent top views of a complete assembly of the device including the means for liquid hair color application. Said top views represent the series of mechanical operations causing the hook to pivot and the color to dispense respectively.
[0066] FIGS. **25**A-**25**B are front perspective views of the complete mechanical arrangement of a preferred embodiment of a hooking applicator with the hook in both the open and closed positions.
[0067] FIGS. **26**A-**26**D are a series of bottom views of a preferred embodiment of the hooking applicator. This series of views depicts four specific points in the process of hair entrainment.
[0068] FIGS. **27**A-**27**B are front views of a preferred embodiment of the hooking applicator in both the open and closed positions.
[0069] FIGS. **28**A-**28**C are full side views of a preferred embodiment of a comb type variation of the hair color variegation device. These views depict the progression of movement of the mechanical assemblies facilitated by a squeezing of the handle of the device.
[0070] FIGS. **29**A-**29**B are perspective views of the front of the preferred comb embodiment of the device. These views are seen from the lower front of the device and depict this embodiment with hair between the teeth. These are views of the device with the hooks in the open and closed positions respectively.
[0071] FIG. **30** is a front view of a cross section of the comb portion of the preferred embodiment.
[0072] FIG. **31** is a perspective view of the front of the preferred embodiment of the comb variation of the device.
[0073] FIG. **32** is a perspective view of the back of the preferred embodiment of the comb variation of the device.
[0074] FIGS. **33**A-B are front left perspective views of the complete mechanical arrangement of a preferred embodiment of a hooking applicator with the hook in both the open and closed positions.
[0075] FIGS. **34**A-B are front left perspective views of a preferred embodiment of the hook and applicator nozzle in both the open and closed positions without rubber facing.
[0076] FIG. **35** shows one embodiment of color spreading elements as part of the saturation chamber.
[0077] FIG. **36** shows another embodiment of color spreading elements as part of the saturation chamber.
[0078] FIG. **37** shows yet another embodiment of color spreading elements as part of the saturation

chamber.

[0079] FIGS. **38**A-D are a series front views of a preferred embodiment of the hooking applicator in functional contact with a parting of hair. These views depict the hook in a progression of hair entrainment positions from the open position to close the closed position.

[0080] FIGS. **39**A-D depict a preferred embodiment of the entire mechanics of the device in front right perspective view. These are a series of views depicting the entire mechanical operation of the device gradually moving from disengaged position to a fully engaged position.

[0081] FIGS. **40**A-B depict an embodiment showing a mechanism for adjusting the size of the entrained section of hair.

[0082] FIG. **41** depicts an entire preferred embodiment of the device in right perspective view including coverings.

[0083] FIGS. **42**A-B depict an entire preferred embodiment of the device in right perspective view including coverings and showing the head hood in both the open and closed position.

[0084] FIGS. **43**A-**43**C show yet another embodiment of the inventive hooking applicator.

[0085] FIGS. **44**A-**44**D show the mechanical operations of the device when dispensing hair color from the hair color container.

[0086] FIGS. **45**A-**45**C show yet another embodiment of the inventive hooking applicator using a different kind of hook for hair color application.

[0087] FIGS. **46**A-**46**D show different views and mechanical actions for the applicator of FIGS. **45**A-**45**C.

[0088] FIGS. **47**A-**47**C show different kinds of hair color containers.

[0089] FIGS. **48**A-**48**B show mechanical action of the applicator of FIGS. **45**A-**45**C.

[0090] FIGS. **49**A-**49**C show still another embodiment of the hooking applicator that uses a worm gear to rotate the hook.

[0091] FIGS. **50**A-**50**E show another embodiment of the device of FIGS. **45**A-**45**C, for coloring a slice of hair.

[0092] FIG. **51**A shows an improvement to the comb device of FIGS. **29**A-**29**B.

DETAILED DESCRIPTION OF THE DRAWINGS

[0093] The present invention is a squeeze operated, hand held device that is used to selectively entrain and color human hair. The invention addresses separate and distinct technical needs of professional hair colorists and individuals desiring a controlled method of selectively coloring their hair.

[0094] The central mechanical aspect of the present invention consists of a hook that pivots along a scalp of hair and entrains a section of hair against a color applicator nozzle. This being the case, it is therefore fitting to begin the detailed description with an explanation of the hook and applicator nozzle as well as the various mechanical interactions thereof in respect to the section of hair that is entrained.

[0095] Relating to the hook and applicator nozzle, the open position′ and closed position′ will be referred to many times. For the sake of minimizing redundancy (see FIGS. **1**A and **1**B), the term “open position” always refers to the hook **2***a* as having pivoted to the farthest position away from the applicator nozzle **11***a*. The term “closed position” always refers to the hollow of the hook **2***a* as having pivoted into contact with the applicator nozzle **11***a* having entrained a section of hair **1***b* between.

[0096] FIGS. **1**A and **1**B depict hooking applicators **1***a* in the open and closed positions respectively. The hook **2***a* in each figure is fixed to an axle **9**; and, the axle **9** pivots in a gear box **6**.

[0097] The hook **2***a* consists of a short length of longitudinally halved tube laterally pivotal on one straight edge and longitudinally tapered on the opposite edge forming the hook point **3***a*.

[0098] Each of FIGS. **2**A, **2**B and **2**C depicts a bottom view of the hooking applicator **1***a* in the closed position; however, FIG. **2**A depicts the hook point **3***a* at the front of the hook **2***a*, FIG. **2**B depicts the hook point **3***a* at the back of the hook **2***a* and FIG. **2**C depicts the hook point **3***a* at the middle of the hook **2***a*. The location of the hook point **3***a* presents a difference in the way the hook **2***a* entrains a section of hair **1***b*: with both types of hooks **2***a* pivoting along a parting of hair **1***c* from the same position relative to the part **1***c*, a hook **2***a* that is pointed on the front **3***a* will entrain less hair than a hook **2***a* that is pointed on the back **3***a*.

[0099] FIG. **3**A depicts the hooking applicator in the open position; more specifically, it outlines the hollow **2***d* (depicted by a bold black line) of the hook **2***a* and depicts the nozzle **11***a* as being partially covered by a layer of viscoelastic foam **11***b* (or any other applicable flexible material). FIG. **3**B illustrates the hollow **2***d* of the hook **2***a* and the cylinder of the applicator nozzle **11***a* are an accurate fit while in the closed position. FIG. **3**C is a sectional view of the closed position and illustrates how this accurate fit becomes a seal as the hollow of the hook **2***a* presses into the layer of viscoelastic foam **11***b* that surrounds the aperture **5** on three sides. The nozzle seal **11***b* prevents leaking of the liquid hair color around the back and sides of the hook **2***a* while in the closed position.

[0100] Referring to FIGS. **1**A and **1**B, notice one foot **4***a* fixed to the front of the applicator nozzle **11***a* and one foot **4***g* fixed to the back of the applicator nozzle **11***a* with the hook **2***a* positioned between. A comfortable contact of the pivoting hook **2***a* with the scalp **1***c* is assured as the hook **2***a* is confined to travel a precise pivotal path between the front foot **4***a* and rear foot **4***g*, and, as the hook point **3***a* is confined to pivot generally flush with the front foot contact point **4***b* and rear foot contact point **4***h*. This mechanical arrangement, therefore, utilizes the front foot contact point **4***b* and rear foot contact point **4***h* as means to allow the hook **2***a* to entrain a section of hair **1***b* while preventing the hook point **3***a* from making forceful contact with the scalp.

[0101] FIGS. **4**A-**4**D bottom views depict another mechanical relationship between the hook **2***a*, front foot **4***a* and rear foot **4***g*, namely, how the hook **2***a*, front foot **4***a*, and rear foot **4***g* function to separate the entrained section of hair **1***b* from the surrounding hair at the scalp as well as to center the entrained section of hair **1***b* within the hollow of the hook **2***a* and maintain the centered position of the section of hair **1***b* through the color coating process. This separation and centering of the entrained section of hair **1***b* occurs as the front contact surface **2***e* and rear contact surface **2***f* of the hook **2***a* slide against the front hook contact surface **4***e* of the front foot **4***a* and rear hook contact surface **4***j* of the rear foot **4***g*, as per a scissor action, while moving from the open to closed position.

[0102] FIG. **4**A depicts the hook **2***a* and applicator nozzle **11***a* in the open position. FIG. **4**B depicts same hooking applicator **1***a* as the hook **2***a* has pivoted toward the closed position enough to have entrained a section of hair **4***b*. FIG. **4**C illustrates the hooking applicator **1***a* as the hook **2***a* has pivoted with the entrained section of hair **1***b* to a point where the hook **2***a* has not quite reached the closed position and the entrained section of hair **1***b* has been pulled in by the hook **2***a* close enough to the applicator nozzle **11***a* for the entrained section **1***b* to have encountered the front foot scissors edge **4***f* and the rear foot scissors edge **4***k*. FIG. **4**D depicts the hooking applicator **1***a* in the closed position with the section of hair **1***b* centered over the aperture **5** of the applicator nozzle **11***a* as well as being centered over the color channel **2***b* and hair channel **2***c* of the hook **2***a*. Also, the entrained section of hair is occupying the front foot channel **4***c* and rear foot channel **4***i*.

[0103] FIG. **1**C depicts a front view of the front foot **4***a* and points to the preferred location of the front scissors edge **4***f* of the foot front **4***a*. As stated above, this scissors edge **4***f* of the front foot **4***a* pushes an entrained section of hair functionally into the closed position. FIG. **1**D depicts this front scissors edge **4***f* located closer to the center of the front foot **4***a* than the preferred location depicted in FIG. **1**C. Relocating this front scissors edge **4***f* relative to the center of the front foot **4***a*, along with relocating the aperture **5**, color channel **2***b* and hair channel **2***c*, so that the said features intersect functionally with the top of the front scissors edge **4***f* in the closed position, changes the amount of hair that is entrained by the individual hooking applicator **1***a* as well as changing the closeness to the scalp of the initial application of color onto the entrained section of hair. Furthermore, if the rotation of the hook **2***a* is not on a particular degree of upswing relative to the bottom of the front scissor edge **4***f*, a portion of the entrained section of hair will be brought against the bottom of the corner of the front foot **4***a* rather than the front scissors edge **4***f* causing strands of hair to become lodged between the hook **2***a* and front foot **4***a*. This will cause the entrained section of hair to become snagged. This front scissors edge **4***f* may occupy various positions relative to the center of the front foot **4***a* and may even be somewhat angled rather than the perpendicular orientation it occupies presently in relation to the bottom of the gear box **6**. In addition to this, the length of the front foot **4***a* may be adjusted in order to change the amount of hair that is entrained. (Note: All of the preceding description of FIGS. **1**C and **1**D also applies to the rear foot which is not visible in said figures. Simply replace the term ‘front’ with the term rear and this will provide the same description of the rear foot.)

[0104] FIG. **3**D is another variation of the feet arrangement featuring a foot bridge **4***d*. This foot bridge **4***d* connects the front foot **4***a* and rear foot **4***g* along the bottom creating one wide foot that surrounds the tip of the hook **3***a* in the closed position. This foot bridge **4***d* creates a further scissor action along the bottom of the hook **2***a*. Furthermore, the hook point **3***a* may be but not necessarily closed on five sides creating a box that is open only to the hook point **3***a* as it pivots toward and establishes the closed position. This foot bridge **4***d* variation is optional.

[0105] View FIG. **3**A-**3**C during the following description of the process by which the entrained section of hair becomes coated with liquid hair color. FIG. **3**A is a front perspective view of the hooking applicator **1***a* showing the hook **2***a* and nozzle **11***a* in the open position with the color channel **2***b* and hair channel **2***c* forming one continuous indentation approximately centered front to back across the hollow **2***d* of the hook **2***a*. FIG. **3**B is a front perspective view of the hooking applicator **1***a* showing the hook **2***a* and nozzle **11***a* in the closed position over a section of hair **1***b* with the color channel **2***b* and hair channel **2***c* positioned approximately centered over the nozzle aperture **5**. FIG. **3**C is a sectional front perspective view of the hooking applicator **1***a*. Said figure shows that the portion of the hollow **2***d* of the hook **2***a* that comes into contact with the applicator nozzle **11***a*, while in the closed position, has flattened the nozzle seal **11***b* that is directly under said portion of the hook **2***a*; however, the color channel **2***b* remains open and, the area of the nozzle seal **11***b* within the hair channel **2***c*, remains raised and fills the hair channel **2***c*. This is because the nozzle seal **11***b* is at least as thick as the hair channel **2***c* is deep. As seen in FIG. **3**B, with a section of hair **1***b* entrained in the closed positon, the entrained section **1***b* is occupying the color channel **2***b* as well as the hair channel **2***c*; furthermore, said hair channel **2***c* is also occupied by a portion of the nozzle seal **11***b* as said portion remains expanded in the hair channel against the entrained section of hair. The portion of the nozzle seal **11***b* that fills the hair channel **2***c* does apply a slight pressure to the section of hair **1***b* entrained therein; however, this pressure is not enough to restrict movement of the entrained section **1***b* through the closed position; the pressure is only enough to prevent the liquid color flowing into the color channel **2***b* from leaking to the outside of the closed position through the hair channel **2***c*. Also, the pressure exerted onto the section of hair **1***b* located within the hair channel **2***c* is such that a desirable amount of tension is maintained on the entrained section **1***b*. This tension allows the device to maintain comfortable control over the entrained sections **1***b* throughout the process.

[0106] While viewing FIG. **3**B, consider a section of hair **1***b* is entrained in the closed position and liquid hair color is exiting the aperture **5**, the liquid fills the color channel **2***b*, thereby surrounding the portion of the entrained section **1***b* that is occupying the color channel **2***b*. As the liquid color continues to exit the aperture **5**, the liquid is prevented by the nozzle seal **11***b* from expanding out from the sides of the hook **2***a* as well as from the back of the hook **2***a* through the hair channel **2***c*. This mechanical arrangement causes the entrained section of hair **1***b* to become coated with liquid hair color **1***d* as well as allows the coated entrained section **1***d* to remain coated as the coated section of hair **1***d* passes through and exits the color channel **2***b*.

[0107] FIG. **3**A depicts a front foot channel **4***c* formed into the side of the front foot **4***a* and rear foot channel **4***i* formed into the side of the rear foot **4***g*. The purpose of each of these indentations is to allow clearance for the entrained section **1***b* of hair to slide through the closed position without getting pinched. As seen in FIG. **3**B, the front foot channel **4***c* has the added benefit of allowing the color coated entrained section **1***d* to pass from the closed position without the color being scraped away from the color coated section of hair **1***d*.

[0108] Each of FIGS. **5**A and **5**B are perspective front views of the hooking applicator in the open and closed positions respectively depicting another color channeling variation featuring a hook **2***a* without a color channel or hair channel. This variation includes an indentation or nozzle color channel **11***d* that is located around the aperture **5** of the applicator nozzle **11***a*. This nozzle color channel **11***d* is open to the front of the closed position as well as the front foot channel **4***c* and will serve to direct the flow of the color coated entrained section of hair **1***d* much the same way as a hook color channel **2***b*. This variation also includes a nozzle hair channel **11***e* located on the applicator nozzle **11***a* behind the nozzle color channel **11***d*. The nozzle hair channel **11***e* opens to the rear foot channel **4***i* in the closed position and is covered by the nozzle seal **11***b* in order to allow the hair to move through the closed position without the risk of color back flow though the nozzle hair channel **11***e*. The nozzle color channel **11***d*, however, is open to the front of the closed position in order to allow the liquid hair color to flow from it.

[0109] FIG. **6**A and FIG. **6**B are front perspective views of the hooking applicator **1***a* depicting another color channeling variation combining both a color channel **2***b* located on the hook **2***a* as well as a nozzle color channel **11***d* located around the aperture **5** of the nozzle, deeper front foot channel **4***c* and nozzle hair channel **11***e*. This variation will provide the most color deposit along the entrained section of hair.

[0110] Considering all of the variations of channeling described above, the shape and dimension of the hook channel **2***b* and nozzle channel **11***d* as well as the size and shape of the aperture **5** will vary according to the viscosity of the liquid hair color as well as the desired degree of control of color flow as well as the size of color bead deposited onto the entrained section of hair **1***b*.

[0111] FIGS. **7**A and **7**B describes a small slender appendage or hook tooth **3***b* extending out from the color channel **2***b* of the hook **2***a*. As seen in FIG. **7**B, the tooth **3***b* extends away from the hollow of the hook **2***a* toward the radial center of the hook **2***a* in such a way that when the hook **2***a* is in the closed position over the applicator nozzle **11***a*, the tooth **3***b* enters into the aperture **5** of the nozzle **11***a*. Since the thickness of the tooth **3***b* is smaller than the dimensions of the aperture **5**, the tooth **3***b* does not obstruct the flow of color from the aperture **5**. In the instance where a pressurized color container is supplying an applicator nozzle **11***a*, a rubber or silicone (or other flexible chemically resistant material) tube gasket **11***c* may be placed snuggly against the inside wall of the applicator nozzle **11***a*. This gasket **11***c* covers the nozzle **11***a* aperture **5** and prevents pressurized as well as non-pressurized color from flowing out.

[0112] Viewing FIG. **7**A and FIG. **7**B in series shows the hook **2***a* and the tooth **3***b* pivoting from the open to the closed position. As the hook **2***a* does this, the tooth **3***b* will push against the portion of the gasket **11***c* located in the nozzle aperture **5**. As the tooth **3***b* pushes against the gasket **11***c*, pressurized color is released. As the hook **2***a* pivots back toward the open position, the tooth **3***b* will exit the aperture **5** and the cylindrical gasket **11***c* will naturally flex back to the closed position over the aperture **5** inside the nozzle **11***a* again blocking the flow of color from the aperture **5**. In this manner, pressurized color may be controlled to flow onto entrained sections of hair only when the hook **2***a* brings the entrained sections to the closed position over the nozzle **11***a*.

[0113] As depicted in FIG. **1**A, the applicator nozzle **11***a* features an aperture **5** as an exit for liquid hair color and a hose **12** functions as a supply line between the color container hoses and the nozzle **11***a*. The nozzle hose **12** extends upward a short distance, perpendicular to the nozzle **11***a* then turns at a right angle, extending back ending in a nozzle hose coupling **13***a*.

[0114] FIGS. **1**A and **1**B depict a preferred embodiment of the device involve a rack and pinion gear arrangement as mechanical means to pivot the hook **2***a*. The hook **2***a* is fixed to the distal front of an axial **9** and the rear portion of the axial **9** pivots within a gear box **6**. A pinion gear **8** is fixed to the portion of the axial **9** contained within the gear box **6**. A rack gear **7***a* pivots the pinion gear **8** within the gear box **6** from underneath.

[0115] As depicted in FIG. **8**, the flexible head **14***a* consists of a straight row of hooking applicators **1***a* connected one to another along the bottoms of the gear boxes **6** by roller couplings **14***d*.

[0116] FIG. **8** also depicts two head mounts **15** attached to the front of the top handle section **20***b*. Each end of the head **14***a* is fixed to the distal front of each head mount **15** forming a head **14***a* attached to a handle **20***a*.

[0117] FIGS. **9**A and **9**B depict the row of hooking applicators **1***a* sharing a single thin flexible rack gear **7***a* that extends along the inside bottom of each gear box **6** with the series of pinion gears **8** seated teeth to teeth into the rack gear **7***a*. Back and forth movement of the rack gear **7***a* causes the pinion gears **8** and therefore the axles **9** and hooks **2***a* to pivot in unison.

[0118] As seen in FIG. **9**A, the head **14***a* is pressed lightly against a parting of hair **1***c* and the head **14***a* flexes into the curve of the scalp. The head **14***a* is placed against the scalp in the upright position thereby allowing each of the front foot contact points **4***b* and rear foot contact points **4***h* (not visible in FIG. **9**) to make functional contact with parting of hair **1***c*.

[0119] FIG. **9**A depicts the device relying on a flat spring **14***c* to allow the head **14***a* to flex. The flat spring **14***c* expands as the head **14***a* flexes into the curve of the scalp.

[0120] The roller coupling **14***d*, shown in FIG. **9**A and FIG. **10**A, is another mechanical feature to aid in the flexing ability of the head **14***a*. A roller coupling **14***d* is fixed to both ends of the head **14***a*; it is a section of tube that fits telescopically over the end of each head mount **15**.

[0121] FIG. **10**A depicts a lip **14***e* formed into each of the head mounts **15** at a location that is as far back from the distal end of the head mount **15** as the head **14***a* is wide. A cap **14***f* is located at the tip of each head mount **15**. Each lip **14***e* and cap **14***f* prevents the roller couplings **14***d* from sliding back and forth along the ends of the head mounts **15**. The inside diameter of the roller couplings **14***d* are slightly larger than the outside diameter of the cylindrical ends of the head mounts, **15** so that the roller couplings **14***d* can freely roll. As seen in FIG. **9**A, each end of the head **14***a* is fixed to each of the roller couplings **14***d* and, as the head **14***a* flexes into the curve of the parting of hair **1***c*, the two hooking applicators **1***a* that are fixed to the roller couplings **14***d* are able to freely pivot over the ends of the head mounts **15**. This pivoting naturally occurs when the head **14***a* flexes against the curve of the scalp and creates a smoother and more complete flexing action.

[0122] The rack slide mount **17**, as viewed in FIG. **8**, is a section of tube that is fixed to the front edge of the top handle section **20***b*. As viewed in FIG. **9**B, one side of an upside down 'L' shaped rod or rack slide **16***a* is positioned snuggly sliding within the tube of the rack slide mount **17**. The other side of the rack slide mount **17** extends straight down then bends out along the side of the head at a slight angle for a short span. The rack slide **16***a* then bends forward and finally tapers down forming the rack gear pin **16***b*. The rack gear pin **16***b* extends directly into a small hole or rack gear seat **7***b* located at the distal end of the rack gear **7***a* forming a snap-in fit between the rack gear pin **16***b* and the rack gear seat **7***b*.

[0123] In order to impart a more complete understanding of the rack slide action, it is necessary to explain in more detail the action of the handle. As seen in FIG. **10**A, the handle **20***a* is composed of a top handle section **20***b*, and a bottom handle section **20***c* connected to one another at the rear of each by a handle hinge **21***a*. This salad tong type configuration is held in the open resting position by the handle hinge spring **21***b* against the handle stop **21***c*. The handle stop **21***c* is a protrusion located on the inside of the bottom handle section **20***c* of the handle hinge **21***a*. As the bottom hinge section **20***c* pivots back toward the open position, the bottom hinge section **20***c* is prevented from opening any further as the handle stop **21***c* comes into contact with the rear bottom edge of the top handle section **20***b*.

[0124] FIG. **10**A-**10**D are side views of the hand of an operator squeezing the handle **20***a* of the device from the open position FIG. **10**A to the closed position FIG. **10**D with FIGS. **10**B and **10**C representing middle handle positions. With the device depicted in FIGS. **10**A-**10**D shown appropriately positioned against a parting of hair **1***c*, one will notice as one views these illustrations in sequence that the top handle section **20***b* (along with the attached head mounts **15** and head **14***a*) remains stationary against a parting of hair **1***c* while the bottom handle section **20***c* is the pivotal section. As such, one will notice, while again viewing these figures in sequence, the visible actuator tine **18** (which is attached to the bottom handle section **20***c*) sliding from the bottom to the top of the head mount **15** and rack slide **16***a*.

[0125] Having established a more complete understanding of the role of the handle as it pertains to the sliding action of the actuator tines against the head mount and rack slide, one may now refer to FIG. **8**. The rack slide actuator tines **18** consist of two rods extending forward from the front edge of the bottom handle section **20***c*. The tines **18** are positioned between the bottoms of the head mounts **15**. The distance between the tine ends **18** is such that the tine **18** on the left is in contact with the left head mount **15** and the tine on the right **18** is in contact with the right head mount **15** as well as the rack slide **16***a*. (Refer to FIGS. **9**A and **9**B for the remainder of the paragraph.) With the actuator tines **18** in this position, squeezing the handle **20***a* will cause the tine **18** on the left to slide upward against the inside of the left head mount **15**, and the tine **18** on the right to simultaneously slide upward against the right head mount as well as the inside of the rack slide **16***a*. As the bottom of the rack slide **16***a* is angled outward (see bold line **16***c*), the upward sliding motion of the right tine **18** against the angled section of the rack slide **16***c* causes the rack slide **16***a* to move outward stabilized by the rack slide mount **17** and head mounts **15** (see FIG. **10**B). As the bottom end of the rack slide **16***a* is attached to the rack gear **7***a* by the rack gear pin **16***b*, the outward sliding motion of the rack slide **16***a* causes the rack gear **7***a* to move to the side. The rack gear **7***a* sliding to the left in this manner causes the hooks **2***a* to pivot toward the closed position over the applicator nozzles **11***a*. In this manner, when the right actuator tine **18** is in contact with the bottom of the rack slide angle **16***c*, the hooks **2***a* are in the open position (see FIG. **9**A). Squeezing the handle until the right tine **18** is at the top of the rack slide angle **16***c* causes the hooks **2***a* to move to the closed position (see FIG. **9**B). Releasing the handle will cause the rack slide **16***a* to return to the inward most resting position against the tension of the rack slide spring **19**. The action caused by a continued squeeze of the handle **20***a* bringing the actuator tines **18** past the top of the rack slide angle **16***c* will be described later in this disclosure.

[0126] The bottom handle section **20***c* (see FIG. **8**) employing only a single tine **18** on the right side against the rack slide **16***a* may also be employed as an alternative embodiment.

[0127] As described in the summary, the present invention features a single squeeze mechanism capable of, in series, entraining the hair and dispensing the liquid hair color onto entrained sections of hair. As described above, engagement of the hooks occurs during the first increment of the squeeze action applied to the handle. The second increment of squeeze action pivots the lever **24***a* so that it pushes up on the level pallet **22**. (See FIG. **8** for a detailed perspective view of the level pallet **22** and lever **24***a*). As both the level pallet **22** and lever **24***a* are hinged to the top of the bottom handle section **20***c*, the upward motion of the bottom handle section **20***c* toward the top handle **20***b* section, combined with the mechanical action of the level pallet **22** and lever **24***a* facilitates the movement of the liquid hair color out of the color container **51***a* and through the channels that direct the color onto the entrained sections of hair

[0128] The following is a detailed description of the second in series mechanical action (dispensing of the hair color) and how this action coordinates with the first action (entraining of hair sections) as the device is in use. The mechanical action will be described while referring to FIGS. **10**A-**10**B. (Note: FIGS. **10**A-**10**B depict side views of the preferred embodiment of the device depicted in FIG. **8**.)

[0129] As seen in FIG. **10**A, an operator functionally holds the device by the handle **20***a* as the device is in the resting position and places the head **14***a* of the device appropriately against a parting of hair **1***c*.

[0130] As seen in FIG. **10**B, the operator squeezes the handle **20***a* causing the bottom handle section **20***c* to lift toward the top handle section **20***b*. Consequently, the tine **18** that is against the rack slide **16***a* moves upward to the top most point of the rack slide angle **16***c* (this point on the rack slide **16***a* appears as a bold square). This causes the hooks **2***a* to be in the closed position over the applicator nozzles **11***a* with entrained stalks of hair **1***b* between. The lifting action of the bottom handle section **20***c* toward the top handle section **20***b* also causes the button contact point **24***c* of the lever **24***a* to contact the lever button **24***d*. This contact causes the lever **24***a* to pivot on the lever hinge **24***e*, thereby, pushing the level pallet contact point **24***b* of the lever **24***a* against the bottom of the level pallet **22**. This, in turn, causes the level pallet **22** to lift toward the bottom of the color container **51***a*.

[0131] FIG. **10**C depicts the handle **20***a* having been squeezed to the point where the actuator tine **18** begins to slide along the section of the rack slide **16***a* that is parallel to the head mount **15**. This allows the hooks **2***a* to remain in the closed position while the level pallet **22** comes into contact with and pushes up on the bottom of the color container **51***a*. The pressure of the level pallet **22** on the color container **51***a* causes the liquid color to begin to move from the color container **51***a* through the color container neck **52***b* and into the manifold intake **53***c*. Continuing through the manifold **53***a*, the liquid color flows through the manifold hoses **53***b*, into the nozzle hoses **12** and nozzles **11***a* and through the color aperture **5** onto the entrained section of hair **1***b*.

[0132] The operator will continue to apply light squeeze pressure to the handle **20***a* while watching for a small bead of color **1***e* to simultaneously form at the front of each hook (see FIG. **9**C). When she sees these color beads **1***e* form she will know that the hair color has exited each nozzle **11***a* aperture **5** and has surrounded the portion of each entrained section **1***b* that is within the closed position. The moment she sees the beads of color **1***e* form, she will maintain the same pressure while slowly pulling the device away from the parting of hair **1***c*. As depicted in FIG. **9**D, she pulls the device away from the parting **1***c*, the constant light pressure on the handle will evenly surround the entrained sections **1***b* with hair color **1***d* as the entrained sections **1***d* pass through the closed position.

[0133] Once the operator has sufficiently coated the entrained sections of hair, she will generally proceed one of two ways: she can release pressure on the handle allowing the device to return to the resting position depicted in FIG. **10**A; this approach allows the coated sections to drop back into the hair. The other option is to release the handle **20***a* only to the point where hair color stops dispensing while maintaining the entrained sections in the closed position. This occurs as the handle **20***a* is released enough for the level pallet **22** to release from the bottom of the color container **51***a* but not enough for the actuator tine **18** to slide down beyond the top of the rack slide angle **16***c*; this mechanical position is depicted in FIG. **10**B (The top of the rack slide angle is depicted as a solid black square located on the rack slide). (The following description of barrier material application does not include correspondent drawings.) At this point, the entrained and coated sections are in a taut and stationary position, extending between the head of the recipient and the head of the device. The operator, while maintaining the entrained sections in this position, and having a free hand, may pick up a folded section of barrier sheet and place it over the entrained sections or perhaps place a section of cotton under the section close to the scalp; any number of barrier material types and techniques known by a person skilled in the art may be applied at this time followed by a controlled release of the barrier treated section into the rest of the hair. Finally, the operator may trace the tip of the parting stem **27** along the scalp, exposing the next parting of hair to be serviced and thereby beginning a new pass of the device along the recipient's hair (see FIG. **10**A for the parting stem **27**). A pass of the device through a recipient's hair, such as the entire pass described above, may be repeated the number of times deemed appropriate by the operator or until the point at which the upward motion of the level pallet **22** onto the color container **51***a* is interrupted by becoming flatly parallel and directly adjacent to the top of the color container housing **26**, thereby flattening and emptying the color container **51***a* (see FIG. **10**D). The color container **51***a* may then be refilled or replaced.

[0134] The following is a description of two types of disposable color containers. These color containers are pre-filled (preferably by a manufacturer), loaded into the device and are discarded when empty.

[0135] Pre-packaged color containers that dispense two part oxidative color or lightener must include a means by which the two reactive components remain separate inside the container until just prior to use. FIG. **11**A depicts an internal container **51***f* within an external container **52***e*, with each container accommodating one of the two hair color components. The internal container **51***f* is filled to capacity so that it is firm. The external container **52***e* is filled but not firm. In addition to the difference in firmness between the two containers, the internal container **51***f* is intentionally manufactured with a structurally weaker front seam and/or weaker plastic film than the external container **52***e*. The difference in firmness in addition to the weak film allows the operator to moderately squeeze this dual container causing the internal container **51***f* to rupture. This rupture releases the color component within, into the other color component contained within the external container **52***e*. The operator will briefly knead the dual container thereby fully mixing the two color components. Also, the rear bottom seam **51***i* of the external container **52***e* and the rear seam **51***i* of the internal container **51***f* are sealed together so that the internal container **51***f* does not float around freely inside the external container **52***e* giving the internal container **51***f* the opportunity to move forward and block the manifold port **51***b* of the external color container **52***e* from the inside. This dual container **52***e*, **51***f* may be discarded once it is empty and replaced by a pre-filled dual container **52***e*, **51***f*. For convenience, the manifold port **51***b* may feature a puncture seal **51***g* adhered to the front. In order to accompany the puncture seal **51***g*, a puncture spike **53***h* will be affixed to the manifold intake **53***c*. This puncture feature allows the operator to mix the components without mess, opening the manifold port **51***b* only at the point where it engages the manifold intake **53***c*.

[0136] FIG. **11**B depicts another preferred dual color container embodiment **52***a*; the purpose of which is to keep the two components of the liquid hair color separate until the two components exit the color container **52***a*. The two (2) components of the liquid hair color are of equal texture and viscosity and are kept separate within the dual color container **52***a* by a barrier **52***b*. The barrier **52***b* essentially forms two separate color containers of equal volume arranged flatly against one another. Each side of the divided color container **52***a* opens to each side of the dual manifold port **52***c*.

[0137] When pressure is applied to this dual color container **52***a*, both hair color components enter each side of the dual manifold port **52***c*. The two components then enter the manifold intake **53***c* where they pass through a section of helical static mixer **52***d* and begin to mix. The partially mixed color then enters the inner tube **53***g* of the manifold **53***a*. The color is further mixed as it passes through the inner manifold tube **53***g* as it also contains a section of static mixer **52***d*. Fully mixed color now exits both ends of the inner manifold tube **53***g* and enters the main outer manifold tube **53***f*, then the nozzle hoses **53***b* and finally exits the nozzle aperture **5**. The operator will proceed with the color service as described previously.

[0138] The following describes the process of reloading the device with color as well as cleaning the various color channels of the device.

[0139] A color container featuring a refill port **51***d* (as seen in FIG. **8**) will not need to be disassembled and can be refilled using a syringe or baster type mixing container with a hollow dispensing stem. The operator mixes the two components of the hair color in the reservoir of the mixing container, secures the lid over the reservoir and injects the mixed color into the color container **51***a* through the refill port **51***d*. Having completed this stage of refilling, the operator secures the lid **51***e* onto the refill port **51***d*.

[0140] Pre-packaged color containers will need to be removed from the device when empty and replaced with one that is full. The following example will be described with a single chamber color container **51***a* (see FIG. **8**), although a dual chamber color container **52***a* could be used for the explanation as well. In order to do this, the operator will release the back of the color container **51***a* from the back of the color container housing **26***a* by disengaging the fastening tabs **51***h* from the fastening pins **26***c* (see FIG. **8** and FIG. **10**A). She will then disengage the manifold mounting bracket **53***e* from the rack slide mount **17**, remove the manifold intake **53***c* from the manifold port **51***b* of the color container **51***a* (see FIG. **8**), bend the manifold **53***a* forward which will disengage the manifold intake **53***c* from the manifold port **51***b* of the color container **51***a*. She will then disengage the manifold port **51***b* of the color container **51***a* from the manifold port bracket **26***b* (see FIGS. **9**A and **10**A) She will then be able to pull the empty color container **51***a* out from the color container housing **26***a*. With the manifold **53***a* still bent forward on the flexibility of the manifold hoses **53***b*, thereby exposing the frontal opening to the color container housing **26***a*, she will then slide a full and sealed color container **51***a* into the opening until the full length of the color container **51***a* occupies the full length of the color container housing **26***a*. Then she will push the manifold port **51***b* of the color container **51***a* onto the manifold port bracket **26***b* in order to secure this port **26***b* as well as the front of the color container **51***a* onto the front of the color container housing **26***a*. Next she will push the fastening tabs **51***h* onto the tab pins **26***c* thereby securing the back of the color container **51***a* to the back of the color container housing **26***a*. Finally, she will urge the manifold intake **53***c* onto the manifold port **51***b* and snap the manifold mounting bracket **53***e* onto the rack slide mount **17**.

[0141] Having refilled the color container **51***a* or, having exchanging an empty single color container **51***a* or dual color container **52***a* with a full one, the operator will now prime the device by squeezing the handle **20***a* until the color exits all of the nozzle **11***a* apertures **5** (See FIGS. **10**A-**10**B and **9**C). The first squeeze with a new color container **51***a*, may cause some color to exit some apertures **5** before others; therefore, the operator will perform this operation over a cleanable surface, paper towel, sink, etc. as the hair color may drip, out of some of the nozzles **11***a* until color is exiting all nozzles **11***a*. The operator will simply wipe the excess color from the nozzles **11***a* with a paper or cloth towel and proceed with the color service.

[0142] In order to minimize the overall number of drawings in this disclosure, the following description of the cleaning procedure does not have supporting illustrations. Refer to FIG. **8** for an approximation.

[0143] In order for the operator to clean the refillable color container **51***a* and manifold **53***a*, she will disengage the color container **51***a* as described above, cap **51***e* the refill port **51***d*, inject water or cleaning fluid into the color container **51***a* through the manifold port **51***b* and place a finger over the manifold port **51***b*. Then she shakes and kneads the color container **51***a* and pours the liquid out of the refill port **51***d* and/or manifold port **51***b*. She will repeat this step until the container **51***a* is clean. In order to clean the manifold **53***a* and nozzles **11***a* she simply engages the color container **51***a* into the device following the reload procedure described earlier, and then fills the container **51***a* through the refill port **51***d* and places the cap **51***e* over the port **51***d* and squeezes the handle **20***a*. Water will jet out of the nozzle **11***a* apertures **5** thereby cleaning the nozzles **11***a* and nozzle hoses **12** as well as all of the hoses and channels of the manifold **53***a*. She may also insert a slender cleaning implement into the various hoses, ports and nozzles during the cleaning procedure.

[0144] Although an operator may rely on disposable color containers **51***a* for regular use, it is advisable for the operator to have a refillable container **51***a* available to fill with water or cleaning fluid in order to utilize the cleaning method just described.

[0145] Other types of color containers may be employed in the device such as a caulk gun type or syringe type arrangement. Also the varied types of containers may be compressed manually, compressed using an electric motor or the color may be dispensed by means of a color container that is under pressure.

[0146] The color containers **51***a*, **52***a* (see FIGS. **8**, **11**A and **11**B) are preferably formed from polyethylene, polypropylene or other type of liquid proof and chemical resistant flexible and easily sealable film. The main tube of the manifold **53***f* (along with the hose couplings **53***d* that are molded into it) is preferably molded from one of a variety of liquid chemical resistant plastic material while the hoses **53***b* may be formed from one of several types of liquid chemical resistant rubber or silicone tubing. The hoses **53***b* may be glued or clamped to the manifold couplings **53***d*; or, all of the couplings **53***d* may be barbed allowing the hoses **53***b* to be removed from the coupling **53***d* yet, attach firmly when in use. The sections of static tube mixer **52***d* will also preferably be of the chemically resistant plastic variety and may be a separate part or formed directly into the inside geometry of the manifold **53***a*. Separate static tube mixers **52***d* may be removable through a threaded cap located on one or both ends of the main manifold tube **53***f*. Removable static tube mixers **52***d* and/or threaded access caps located on the ends of the main manifold tube **53***f* are features that make the manifold **53***a* easier to clean. Also, the sections of static tube mixer **52***d* may also be located within the manifold hoses **52***b*. Alternatively, the entire geometry of the non-mixing manifold **53***a*, including the hoses **53***b* and manifold intake **53***c* may be molded as one part from a liquid chemical resistant rubber or silicone.

[0147] When considering the functionality of the manifold **53***a*, notice the L-shaped manifold hose **53***b*. This L-shape provides a corner that acts as a weak leverage point and allows the pressurized liquid filled hose **53***b* to bend easily as the head **14***a* of the device conforms to the curve of the scalp.

[0148] Another unique feature of the device is a mechanical arrangement that gives the operator the ability to stop the flow of color to individual applicator nozzles while allowing other applicator nozzles to flow. The mechanism effectively pinches a hose closed with the push of a lever.

[0149] As seen in FIG. **12**A-**12**C, each end of the rigid plastic manifold tube **53***f* is detachably affixed against the top of each of the head mounts **15** by a manifold mounting bracket **53***i*; one additional manifold mounting bracket **53***e* extends from the top center of the main manifold tube **53***f* and attaches to the rack slide mount **17**. Fixed along the length of the manifold tube **53***f* are several short lengths of rigid tube that function as couplings **53***d* for the lengths of hose **53***b* that extend away from the manifold tube **53***f*. Fixed to the front of the manifold tube **53***f* are clamp lever mounting brackets **54***e*; one above each of the hose couplings **53***d*. Attached pivotal to each of the lever mounting brackets **54***e* is a clamp lever **54***a*. The top of the clamp lever **54***a* extends back across the top front of the handle a short distance and at a slight angle while in the resting position. This top section of the clamp lever **54***a* is flat and serves as a thumb contact **54***b*. The bottom section of the clamp lever **54***a* extends straight down to a point just below the bottom of the hose couplings **53***d*. At this point the clamp lever **54***a* makes a sharp angle back to a point where it has extended slightly behind the bottom of the hose coupling **53***d*. Now this bottom end of the clamp lever **54***a* makes a final sharp turn and crosses the back of the hose slightly below the hose coupling **53***d* forming the clamp lever hose contact **54***c*. This being the shape of the clamp lever **54***a*, when an operator places a thumb onto the thumb contact **54***b* and presses down, the bottom of the clamp lever hose contact **54***c* moves forward against the hose **53***b* just below the point where the hose **53***b* attaches to the coupling **53***d*. As the operator continues to press on the thumb contact **54***b*, the clamp lever hose contact **54***c* pinches the hose **53***b* forward against the pinch plate **54***f* thereby stopping the flow of color through that hose **53***b* (see FIGS. **12**B and **12**C side views depicting the hose clamping mechanism in the disengaged and engaged positions respectively). Once the operator presses the thumb contact **54***b* down to the farthest point, two interlocking hooks **54***d*, one on the bottom of the thumb contact **54***b* and one on the front top of the head mount **15**, will lock together thereby holding the clamp lever **54***a* in the hose pinching position. The operator simply needs to move the thumb contact slightly to the side and the clamp lever lock **54***d* disengages restoring color flow to the tube **53***b*.

[0150] FIGS. **12**A-**12**C depict a necessary variation of the actuator tines. Since it is chosen, although not necessary, to have all of the parts of the present hose clamping mechanism built onto and around the manifold in such a manner that the manifold in the present embodiment sits lower on the head mounts than in similar embodiments described; including the actuator tines, as they have previously been arranged, into the present embodiment will cause the actuator tines to run into the manifold before they have a chance to slide the functionally necessary distance up the length of the head mounts and rack slide. Therefore, depicted here are telescopic actuator tines. As viewed in FIG. **12**A, the slide tine on the other side of the device, although not visible, will have all of the features of the visible slide actuator tine described in the following:

[0151] The rear end of the slide actuator tine **93** is within in a slide tine seat **94**. The front end of the slide actuator tine **93** has, fixed and extending away perpendicular to the outside, a slide actuator tine channel pin **96**. This channel pin **96** is seated within a channel **95** formed into the head mount **15**; said channel **95** extends the entire length of said head mount **15** and is open to the inside.

[0152] The bottom handle section **20***c*, as well as the slide tine seat **94** that is fixed to it, as seen in FIG. **12**B, are farther away than said parts of FIG. **12**C. As the front of the bottom handle section **20***c* moves closer to the top handle section **20***b*, the front of the bottom handle section **20***c* also moves closer to the head mounts **15**. This is why fixed actuator tines eventually run into the main manifold tube. The sliding actuator tine **93** overcomes this problem. FIG. **12**B shows the front of the slide tine **93**, with the fixed channel pin **96** seated inside the tine channel **95**, (the channel pin **96** is seen as a bold dot) fully extended from the tine seat **94**. FIG. **12**C shows that, as the bottom handle section **20***c* moves up and gets closer to the head mounts **15**, the tine channel pin **96** follows the tine channel **95** and causes the tine seat **94** to move forward over the slide tine **93**. This arrangement allows the front of the tine **93** to track the length of the head mount **15** thereby remaining in the same position relative to it.

[0153] This novel actuator tine arrangement may be included in any embodiment of the device that requires actuator tines.

[0154] Another multi-hooking mechanism device embodiment of the device allows the hooking mechanisms to be positioned closer together than the multi-hooking mechanism device embodiment described previously. The previously described embodiment discloses a row of hooking applicators that are positioned side by side in such a way that the pivoting motion of the hooks are parallel to the line represented by the row of hooking applicators. This means that the more open the hook is relative to the applicator nozzle, the farther away the individual hooking applicators must be from one another. This is because the hook can only open so far as the point at which the hook makes contact with the applicator nozzle of the neighboring hooking applicator. The closer the neighboring hooking applicator, the less the hook can open. Another solution to this problem is to make the hooks smaller. This however may not be a desirable solution as this may cause the sections of hair that are entrained to be smaller than desired.

[0155] To overcome this shortcoming, the present embodiment features a row of hooking mechanisms that are at an angle to one another so that when each hook is in the open position, each hook is positioned, in front of each neighboring applicator nozzle; therefore, each hook does not bump into each neighboring applicator nozzle. One way to accomplish this is depicted in FIG. **13**. This figure depicts a top view of a row of hooking applicators **1***a* arranged side by side and angled as described above. This row of hooking applicators **1***a* is arranged as a device head **14***a*; yet, this head **14***a* is depicted without the rest of the device. The rest of the device is omitted as no further mechanical change is required of the device in order to accommodate the head **14***a* arrangement described below. FIG. **13** shows the tops of the gear box **6** cut away to expose a rack gear **7***c* with angled teeth **7***d* seated against an angled pinion gear **8** arrangement. In addition to depicting the angled pinion gear **8** position and angled rack **7***a* gear teeth **7***d*, FIG. **13** also depicts the hooks **2***a* in the open position in front of the adjacent nozzle **11***a* rather than against the nozzle **11***a* as per the previously described device head **14***a* arrangement; therefore, this angled hooking applicator **1***a* embodiment solves the above stated shortcoming by allowing the hooks **2***a* to remain the same size while positioning the hooking applicators **1***a* closer together.

[0156] As an alternative to the previously described rack and pinion gear means, the following describes an embodiment of the device that utilizes a mechanical lever action as a means to pivot the hooks. As per the device head embodiment described immediately prior, this device head embodiment likewise features a series of hooking applicators that are arranged in an angled configuration so that the hook axles are at an angle relative to the parting of hair, thereby, allowing the hooks to pivot in front of the adjacent nozzles. Although the present embodiment features hooking applicators that are arranged in said manner, this lever action embodiment may also be arranged such that the pivotal relation of the hook axles to the parting may also be approximately perpendicular as per the first device head configuration described in this disclosure.

[0157] It is necessary to state the following at this time; the many parts of the device that are not mentioned in the following description will be assumed to function in like manner to the first embodiment of the device described in this disclosure. This, in order to avoid redundant descriptions.

[0158] As seen in the two different angles of front view (FIGS. **14**A and **14**B) depicting the lever action hooking applicator **70**, the present embodiment features a hooking applicator **70** with a hook **2***a* that pivots on a hook seat **68**, said hook seat **68** being located at the top front of the hooking applicator body **69**. The hook **2***a* features a lever **65** that extends away from the back of the hook **2***a*. The hook **2***a* pivots as the hook slide **61**, and therefore, the hook slide tip **63** slides forward, guided within the hook slide bracket **64**. As the hook slide **61** moves forward, it slides underneath the hook lever **65** causing the lever **65** and therefore the hook **2***a* to pivot. The hook slide tip **63** will slide forward against the hook lever **65** until the hook **2***a* closes over the nozzle **11***a*. Conversely, as the hook slide **61** backs away from the hook lever **65**, the hook **2***a* pivots back to the open position, pulled as such by the tension of the hook spring **66**.

[0159] Having described the mechanical action of the individual lever action hooking applicator **70**, the following is the series of mechanical actions that occur in order to simultaneously pivot all of said type of hooks along a device head comprised of multiple lever action hooking mechanisms.

[0160] As viewed in FIGS. **15**A and **15**B, the handle **20***a* of the device compresses and the actuator tines **18** begin to slide up against the angled bottom section **55***b* of each slide rack lever **55***a*. (The angled bottoms of the slide rack levers **55***a* are indicated in FIG. **15**A by two bold black lines.) As the actuator tines **18** continue to slide upward against the rack slide lever angles **55***b*, the slide rack levers **55***a* begin to close against the head mounts **15**. As seen in FIG. **15**B, the inward closure of the slide rack levers **55***a* against the head mounts **15** cause the slide rack actuators **56** (which are fixed to the outside of the slide rack levers **55***a*) to begin to enter the actuator channels **57**; these channels **57** are openings located on the slide rack mounts **58**. Each slide rack mount **58** is fixed to each head mount **15**. As the slide rack actuators **56** continue to enter the actuator channels **57**, the angled fronts of the slide rack actuators **56** cause the slide rack seats **59** and the slide rack **60***a* on which they are attached to move forward.

[0161] Sandwiched between the two slide rack plates **60***b* of the slide rack **60***a* are the top sections **62** of the hook slides **61**. Continuing to view FIG. **15**B, the bottom sections of these hook slides **61** are fixed to and extend forward perpendicular to the top sections **62** forming the ‘L’ shaped hook slide **61**. The hook slide tips **63** move back and forth in the hook slide seats **64**. While the rack slide **60***a* moves forward, the top sections of the ‘L’ shaped hook slides **62** and therefore the hook slide tips **63** also begin to move forward. As the hook slide tips **63** move forward within the hook slide brackets **64**, the hook slide tips **63** push forward on the hook levers **65** causing the hooks **2***a* to move from the open to the closed position. As the operator releases the handle **20***a* the above mechanical process reverses, the hooks **2***a* return to the open position by the tension of the hook springs **66** and the rack slide **60***a* returns to the resting position by the tension of the rack slide spring **67**.

[0162] The next mechanical operation of this lever action hooking applicator embodiment to be described is the curvature conformation feature (refer to FIGS. **16**A and **16**B). As described above, the mechanical relationship between the slide rack **60***a* and top sections of the hook slides **62** are responsible for the pivoting action of the hook **2***a*. In addition to this function, the top sections of the hook slides **62** and slide rack **60***a* also give the device the ability to conform to the curve of the head. In order for the head **14***a* of the device to curve, the individual hooking mechanisms **70** must be able to move up and down a short distance relative to the head mounts **15** as well as pivot to the side slightly. The contiguous way in which the top sections of the hook slides **62** are positioned within the slide rack **60***a* allow the top portions of the hook slides **62** to move up and down as well as pivot side to side radially. Now, when the head **14***a* of the device is urged against the scalp, each hooking applicator **70** moves from the resting position to the position it must assume in order for it to cooperate with the other attached hooking mechanisms **70** in assuming the particular degree of curvature. As each hooking applicator **70** moves, so does the top portion of each hook slide **62** sandwiched within the slide rack **60***a*. Now as the top portion of each hook slide **62** changes position pivotally from side to side as well as up and down differently from the other top portions of the hook slides **62**, they do not change position pivotally from front to back as the slide rack **60***a* prevents this front to back pivoting. So, the slide rack **60***a* can move forward and back, thereby causing the hooks **2***a* to pivot from the open to closed position in unison even as the individual hooking mechanisms **70** pivot from side to side as well as move up and down differently from one another.

[0163] The final difference that will be described is a variation of position and shape of the color manifold. Considering an embodiment of the device which employs a rack and pinion gear arrangement to pivot the hooks, this gear driven embodiment eliminates the option of positioning the manifold hoses so that they extend from the manifold directly to the applicator nozzles through the area where the rack and pinion gears are positioned thereby eliminating the applicator hose. A mechanical arrangement that allows the manifold hoses to run straight to the back of the applicator nozzle renders a device head with less plumbing and therefore easier cleaning. In addition to a head with less plumbing, the manifold hoses can be shorter, and therefore, take up less space.

[0164] As seen in FIG. **17**, the present lever action hook embodiment features a low manifold color container **52***f* with a single hose **52***g* (see also FIG. **15**A) that extends from the front, extending down to the low manifold **52***h* located level with and in back of the applicator nozzles **11***a*. Also, the low manifold hoses **52***i* extend forward away from the manifold **52***h* a short distance and connect to the applicator nozzles **11***a*.

[0165] Hose connectors as well as any other pertinent part not described in this low manifold color container **52***h* arrangement may be adapted to here from previously described color container arrangements.

[0166] Any embodiment of the hooking applicator may substitute liquid color application onto entrained sections of hair for the application of hair color chalk, mascara or any other type of hair color or hair treatment that can be formed into a solid or semi solid stick. As depicted in FIG. **18**, this embodiment of the hooking applicator **71** features a spring **73** loaded tube **72***a* that is positioned open end **72***b* down between the front foot **4***a* and rear foot **4***g* of the hooking applicator **71** so that the open end **72***b* of the tube **72***a* will be centered within the hollow of the hook **2***a* in the closed position. This tube **72***a* is positioned in such a way that it may be fixed or detachable to the front foot **4***a* and rear foot **4***g*. If it is detachable, the spring **73** loaded tube **72***a* will have a tab **74***b* fixed to opposite sides of the tube's open end **72***b*. The upper portion of the front foot **4***a* and rear foot **4***g* will have a tab seat **74***a* indented centered on the upper inside. Now, the tabs **74***b* of the spring loaded tube **72***a* will snap securely into and out of the tab seats **74***a*. A stick of hair treatment **75** is positioned between the compressed spring **73** and the bottom of the tube **72***b*. The stick of hair treatment **75** is held from springing out of the opened end of the tube **72***b* by two flexible, thin, intersecting cross members **76**. These cross members **76** are attached to the open end of the tube **72***b* and intersect at or near the center of the opening of the tube **72***b*. Alternatively, the cross members **76** may be substituted for one or more tiny flexible tabs attached to the edge of the tube open end **72***b* in such a way that they face toward the center of the tube open end **72***b* and may or may not connect as they may radiate only partially toward the center.

[0167] The viscosity or hardness of the stick of hair treatment **75** must be such that it is soft enough to wear away easily from the stick **75** onto the section of hair that passes over the exposed tip **77** of said stick **75** yet the stick of hair treatment **75** must be of the viscosity or hard enough so that, as the stick of hair treatment **75** is being pushed against the cross members **73** by the spring, the stick **75** will not extrude through the cross members **76** while the device is not in use.

[0168] The hollow of the hook, features an indentation **2***b* that is the diameter and shape of the section of the rounded tip **77** of the stick of hair treatment **75** that is protruding from the open end **72***b* of the tube **72***a*. Now, as the hook **2***a* closes over the tip **77** of the stick of hair treatment **75**, the tip of the stick **77** seats accurately into the indentation **2***b* in the hollow if the hook. As the hook **2***a* entrains a section of hair, the front foot **4***a* and rear foot **4***g* center the entrained section of hair over the indentation **2***b* located in the hollow of the hook **2***a*. Once the hook **2***a* has closed over the tip **77** of the stick of hair treatment **75** with the entrained section of hair, the hair will move through the closed position and will be coated with the hair treatment. As the tip **77** of the stick of hair treatment **75** wears away with repeated runs of entrained sections of hair it will be continually fed to the tip **72***b* of the tube **72***a* against the tension of the spring **73**. The intersecting cross members **76** hold the tip **77** of the hair treatment stick **75** in place at the end of the tube **72***b* and allows the tip **77** of the stick hair treatment **75** to wear away evenly as the cross members **76** are able to move slightly during repeated runs preventing un-worn away ridges from forming on the tip **77** of the stick of hair treatment **75** directly under the cross members **76**.

[0169] The operator will feel the need to adapt the way she uses the device to better accommodate the various needs and requests of the patrons seeking hair color variegation services. The operator has the option to vary the distance between the rows of color treated hair. This allows the recipient to choose within a range of more or less color treated sections placed in the overall color service. In addition to this, the recipient may choose within a range of thick or fine individual color treated sections. If the operator places the head of the device close to a parting of hair, the device will entrain and therefore treat finer sections of hair. The farther away the operator places the head of the device from the parting of hair the thicker the entrained and treated section will be. While keeping the head of the device parallel to the parting, the operator may also slightly stagger the successive placements of treated rows from side to side. By adjusting the three technical variables described above, various aspects of the final appearance of the color service may be changed by using a single device head. However, far more variation in the final appearance is possible with a device that has multiple device heads to choose from. Detachable and interchangeable device head embodiments will now be described and illustrated.

[0170] A wide range of varied head types may be embodied by creating a range of hook/hooking applicator sizes and arranging them at various distances from one another onto heads of different widths. A larger hook/hooking applicator will entrain a wider/larger section of hair and, conversely, a smaller hook/hooking applicator will entrain a narrower/smaller section of hair. Also, a head with hooks that are spaced farther apart or closer together will render each entrained section along the row of entrained sections farther apart or closer together from one another. Obviously then, a wider head will render a wider row of entrained sections.

[0171] FIG. **19**A depicts a device head **14***a* with three larger hooks **2***a* arranged at a greater distance from one another comprising a head **14***a* of perhaps medium width. FIG. **19**B depicts a device head **14***a* with five smaller hooks **2***a* arranged at a closer distance to one another forming a head **14***a* of perhaps medium width. FIG. **19**C is also perhaps a medium width device head **14***a* with four larger hooks **2***a* arranged closer to one another. FIG. **19**D is a device head **14***a* of three larger hooks **2***a* positioned close to one another comprising a head **14***a* of narrower width. FIG. **19**E depicts six smaller hooks **2***a* positioned close to one another along a wide head **14***a*. There are many more head variations possible and may it suffice to state that all will occur as obvious in light of what has thus far been disclosed.

[0172] There are numerous mechanical arrangements that may be employed to create a head that quickly and easily detaches and reattaches to the body of the device. One preferred embodiment of the detachable head is depicted in FIGS. **20**A and **20**B. These figures describe a head **14***a* that includes roller couplings **14***d* that pull away from the head mounts **15** as well as a manifold intake **53***b* that pulls away from the color container coupling **51***b* and a rack gear pin **16***b* that pulls away from rack slide seat **7***b*. Re-attaching the head in this instance simply requires the operator to re-attach what has been detached.

[0173] Another detachable head embodiment is depicted in FIG. **21**A-**21**C. These figures describe an embodiment of the device featuring a detachable arrangement where the head **14***a* of the device as well as the head mounts **15** and actuator tines **18** detach. This arrangement allows the width of the head **14***a* to vary from one detachable head to another. The FIG. **20** detachable head arrangement alone does not.

[0174] The FIG. **21**A-**21**C embodiment utilizes small spring loaded release levers (**78***a* and **78***h*). FIG. **21**C is included in order to provide a magnified view of the type of release lever (**78***a* and **78***h*) used. The head release lever **78***a* allows the front portion of the top hinge plate **20***b* to detach. In this manner, the head **14***a*, head mounts **15**, rack slide mount **17** and rack slide **16***a* detach from the device with one press of the head release lever **78***a* trigger **78***b*. An actuator release lever **78***h* allows the front portion of the bottom hinge plate **20***c* and therefore the actuator tines **18** to detach. Actuator tines **18** that are detachable are necessary because a wider head mount **15** requires actuator tines **18** that are wider.

[0175] As mentioned above, a head release lever **78***a* is positioned on the front of the top handle section **20***b*. The trigger **78***b* side of the lever **78***a* is curved down following the contour of the distal front of the top handle section **20***b* and the latch side **78***c* extends straight back and then ends at a short right angle bend forming the latch pin **78***d*. In the resting position, the latch pin **78***d* rests in a small hole or latch pin eyelet **78***e*. The eyelet **78***e* opens on the inside to the hollow insert seat **79** of the detachable front of the top handle section **20***b*. The insert seat **79** of the detachable front of the top handle section **20***b* is open at the back and fits over the insert tab **80***a* attached to the distal front of the top handle section **20***b* as it is in the detached state. An indentation or pin seat **80***b* is located on the top surface of the insert tab **80***a*. The pin seat **80***b* lines up with the pin eyelet **78***e* when the insert tab **80***a* is the fully engaged position over the insert seat **79**. This allows the latch pin **78***d* to seat through the eyelet **78***e* and into the pin seat **80***b* thereby locking the detachable front of the device onto the body of the device with the tension of the elbow type latch spring **78***f* holding the latch pin **78***d* in the pin seat **80***b*. In order to detach the front of the device from the body, the operator simply presses down on the trigger **78***b* of the lever **78***a* causing the trigger side **78***b* and latch side **78***c* to pivot on the hinge **78***g*. As the trigger side **78***b* of the lever **78***a* pivots down, the latch pin **78***d* pivots up and out of the pin seat **80***b* against the tension of the latch spring **78***f*. With the latch pin **78***d* lifted out of the pin seat **80***b*, the operator simply pulls forward on the detachable front of the device and it simply slides off (FIG. **21**B depicts the detachable head in the detached position). As the latch pin **78***d* is angled on the back, the operator simply slides the insert seat **79** over the insert tab **80***a* and the latch pin **78***d* lifts as it slides over the tab **80***a* and then clicks down into place within the latch pin seat **80***b* urged by the tension of the latch spring **78***f*.

[0176] Also depicted in FIGS. **21**A and **21**B, the actuator tines **18** detach from and reattach to the front of the bottom handle section **20***c* utilizing the actuator release lever **78***h*. See the description of the action of head release lever **78***a* above for the action of the actuator release lever **78***h*.

[0177] Another preferred detachable head embodiment is depicted in FIG. **22**A-**22**F. This embodiment utilizes a dial with a spiral thread to adjust the width of both the head mounts and the rack slide actuator. The dial adjustable head mounts will be described first followed by a description of the dial adjustable rack slide actuator. It is important to note while considering the following dial adjustable head mount mechanical arrangement that the rack slide **16***a*, rack slide mount **17** and rack slide spring **19** are appropriately attached to one of the head mounts.

[0178] FIG. **22**A shows each of the two head mounts **15** consists of a slide plate **81** and a head mount **15**. Each slide plate **81** is mounted separately onto the front of the top handle section **20***b*.

Fixed to the top front of the top handle section **20***b* are two slide rails **82** that run parallel to and are a short distance from one another. Each slide plate **81** has two slide rail fittings **83** formed into it. Each fitting **83** tightly surrounds each slide rail **82** on three sides but not so tight as to prevent each fitting **83** from sliding along each rail **82**. This slide arrangement confines the movement of each slide plate **81** as well as the head mount **15** fixed to it to a side to side slide. FIGS. **22**A and **22**B show a single dial **84** is positioned over both slide plates **81** as they are fitted onto both slide rails **82**. The dial **84** is mounted over the slide plates **81** by an axle **85** that is fixed to the top handle section **20***b*. The dial **84** is positioned in such a way that the dial **84** presses down firmly onto the rail fittings **83** yet the dial **84** can turn. A knob 86 or more preferably a key slot **86** will be positioned at the center of the dial **84** so that the operator can easily turn the dial **84**. A key slot **86** is more preferable because the dial **84** is also the thumb rest for the operator; therefore, a key slot **86** will be less obstructive for this purpose. The device will also include a key that is similar in dimension to a coin so that the operator may also use a coin to turn the dial **84**.

[0179] Radiating from the axle **85** along the bottom of the dial **84** to the outside edge of the dial **84** are two threads **87**. These threads **87** are curved thin grooves that form a spiral across the bottom of the dial **84**. Fixed to and protruding from the top surface of each slide plate **81** is a short small and perhaps cylinder thread insert **88**. As the name implies, the thread insert **88** seats into the thread **87** of the dial **84**. Now, as the operator turns the dial **84**, each thread insert **88** will move back and forth along each thread **87** in turn causing each slide plate **81** and head mount **15** to slide back and forth along each slide rail **82**.

[0180] There are numbered dial positions **89** aligned with the back of the dial **84**. The area where the numbers are located is raised to the same level as the slide rail fittings **83**. There may be any number of dial positions **89** indicated but preferably the number of positions will be the same as the number of head widths available to the device. Our preferred dial embodiment has three positions.

[0181] A setting indicator **90** mark is positioned on the back edge of the top surface of the dial **84**. Positioned on the bottom of the dial **84** directly under the setting indicator **90** is a small protrusion **91**. Positioned along the raised numbered area of the top handle section **20***b* are indentations **92**. There is one indentation **92** positioned under the dial **84** directly in front of each dial position **89**. Also, each indentation **92** is in line with the dial protrusion **91**; so, as the operator turns the dial **84**, the protrusion **91** will snap into the indentations **92**. Each snap-in, numbered dial position **89** corresponds to a specific width of a particular detachable head.

[0182] FIGS. **22**B, **22**C and **22**D are all depictions of three dial **84** positions as well as each corresponding head mount **15** position. FIG. **22**E depicts the dial adjustable head mount embodiment including the head **14***a*.

[0183] FIG. **22**F depicts a bottom view of the device showing the actuator tines **18***a* with the same dial **84** controlled adjustability feature as the head mounts **15**. Each of the two actuator tines **18***a* consists of a slide plate **81** and an actuator tine **18***a*. Each slide plate **81** is mounted separately onto the bottom front of the bottom handle section **20***c*. Fixed to the bottom front of the bottom handle section **20***c* are two rails **82** that run parallel to and are a short distance from one another. All other mechanical aspects of the dial controlled adjustability feature of the actuator tines are identical to the mechanical aspects of the dial controlled adjustability feature of the head mounts described previously.

[0184] FIG. **23**A depicts a preferred pen or marker type embodiment of the device that is more compact and less complicated to use than the previous embodiments. This embodiment may be the most likely, of all of the embodiments presented so far, to be directed to the consumer market as it features only a single hooking applicator **1***a* making it more possible for consumers to use the such a device on one another; or, on him or herself.

[0185] In general, FIG. **23**A depicts a single hooking applicator or head **28** of the device fixed to a body plate **50***a* along the side of the gear box **6**. Extending away from the rear of the body plate is a parting stem **27**. A squeeze plate **44**, approximately the same dimensions as the body plate **50***a*, is

positioned a distance from and face to face to the body plate **50***a*.
[0186] Although the present embodiment features a single hooking applicator, this embodiment may also maintain nearly the same ease of use and mechanical configuration while featuring two or more hooking applicators as the head of the device. For instance, the present embodiment may feature a head comprised of two or three hooking applicators that are joined to one another level and side by side. This head may also have a body plate fixed to the side of one of the gear boxes, and so on, including all of the mechanical features described in the following. Furthermore, like the single hooking applicator head, this two or three hooking applicator head does not require a head conformation feature as does the four or more hooking applicator head described previously. This is because the span of the two or three hooking applicator head is narrow enough that a certain fixed orientation of said head will overcome the need for the head to bend or flex into the varied curvature of the scalp.
[0187] An illustrated description of the hooking applicator has been presented earlier in this disclosure; therefore, a description of the hooking applicator in the following will occur in a cursory manner in order to coordinate it with the detailed illustrated description of the mechanisms involved in hook engagement and liquid hair color discharge.
[0188] The present single hooking applicator head embodiment features a similar sequential hair entraining and color dispensing trigger function to the previously described multiple hooking applicator head embodiment. The mechanism responsible for this will be described below.
[0189] As seen in FIG. **23**B, a single trigger **29***a* is hinged **30** to the front of the body plate **50***a* to both body plate wings **50***b*. As the trigger **29***a* moves from the open resting position, it does so against the resistance of the trigger spring **32**. The wound pivotal section of the spring **32** is positioned with the trigger hinge pin **31** running through it. The trigger spring **32** is leveraged on one end to the trigger **29** and on the other end to the top body plate wing **50***b* by the trigger spring eyelet **33**.
[0190] As seen in FIG. **23**C, the trigger **29***a* pivots on the trigger hinge **30** toward the squeeze plate **44** causing the hook **2***a* to pivot from the open toward the closed position facilitated by a series of coordinated lever and slide hinge mechanisms that originate at the trigger **29***a*. The following is a detailed description of this mechanical operation.
[0191] While viewing FIGS. **23**B and **23**C in sequence (also, see FIGS. **24**A and **24**B in sequence), the trigger **29***a* pivots from the resting position toward the squeeze plate **44** causing the trigger slide **35** to pivot on the trigger slide hinge **36***a* while being pushed forward on the trigger slide hinge **36***a* by the trigger slide push rod **36***b*. This slide forward of the hinged back of the trigger slide **35** occurs as the push rod **36***b* is hinged to both the trigger slide **35** as well as the trigger slide hinge **36***a* on one side while being hinged to the back of the body plate **50***a* by the squeeze plate stabilizer hinge **46** on the other side; therefore, it is the coordinated hinged slide lever action of the push rod **36***b* and the trigger **29***a* that pushes the trigger slide **35** at an angle forward through the trigger slide guide **37**.
[0192] As the trigger slide **35** moves forward, sandwiched between the trigger slide guide **37** and the top wing **50***b* of the body plate **50***a*, the front of the trigger slide **35** is confined to a specific angled forward path as the trigger slide channel **39** moves with the fixed trigger slide guide pin **38** positioned within. As the distal front of the trigger slide moves forward, it encounters the contact angle **40***b* (indicated by the single short bold line) of the slide wedge **40***a*. The movement of the front of the trigger slide **35** over the slide wedge contact angle **40***b* causes the slide wedge **40***a* to slide downward against the tension of the slide wedge spring **40***d*, guided as it is sandwiched between the body plate **50***a* and the slide wedge bracket **41**. This downward motion of the slide wedge **40***a* causes the wedge section **40***c* to wedge between the front of the body plate **50***a* and the rack gear slide **42***a*. As the rack gear slide **42***a* is fixed to the rack gear **7***a*, the rack gear **7***a* slides to the side. This motion of the rack gear **7***a* causes the pinion gear **8** to turn thereby bringing the hook **2***a* to the closed position over the applicator nozzle **11***a*. Squeezing the trigger **29***a* to the point

where the hook **2***a* becomes closed over the applicator nozzle **11***a* causes the trigger squeeze plate contact point **29***b* to come into contact with the squeeze plate **44**. Continuing to squeeze the trigger **29***a* maintains the hook **2***a* in the closed position as the trigger slide **35** simply continues to move forward over the fully engaged slide wedge contact angle **40***b*, while the trigger squeeze plate contact point **29***b* continues to push the squeeze plate **44** toward the body plate **50***a*.

[0193] As seen in FIG. **23**B, the movement of the squeeze plate **44** is confined to a face to face approach toward the body plate **50***a* by two (2) slide mechanisms: a squeeze plate slide hinge **48***a* positioned at the front of the body plate **50***a* and a squeeze plate slide stabilizer **45** positioned at the back of the body plate **50***a*. The squeeze plate slide hinge **48***a* guides this end of the squeeze plate **44** to slide back and forth face to face toward the body plate **50***a* along the squeeze plate slide hinge channels **49***a*. The squeeze plate slide stabilizer **45** also guides the movement of the squeeze plate **44** to a back and forth face to face slide at the back of the squeeze plate **44**; however, this rectangular shaped rod **45**, as it is hinged to the body plate **50***a* on one side and hinged slidable to the squeeze plate **44** on the other side within the squeeze plate stabilizer slide brackets **47**, allows the squeeze plate **44** to approach the body plate **50***a* along the same axis (x) as the trigger **29***a* pivots with little wobbling edge to edge along the y axis.

[0194] As seen in FIGS. **24**A-**24**D, the approach of the squeeze plate **44** toward the body plate **50***a* occurs against the tension of the dual elbow squeeze plate spring **48***b* located along the bottom front of the device (In order to provide further clarity, the dual elbow spring **48***b*, although it is positioned along the bottom of the device, is shown in bold black in FIG. **4**A). As the squeeze plate **44** approaches the body plate **50***a* it does so preferably at an angle back relative to the body plate **50***a* so that the back of the squeeze plate **44** comes into contact with the back of the body plate **50***a* first (as seen in FIG. **24**C) followed by an angled forward approach of the front of the squeeze plate **44** toward the front of the body plate **50***a* until the full length of the squeeze plate **44** is in full face to face contact with the body plate **50***a* (as seen in FIG. **24**D). This approach of the squeeze plate **44** toward the body plate **50***a* is preferred in order that, when a full color container **51***j*, such as the preferred type depicted in FIG. **24**A, is loaded functionally into the device, the color container **51***j* is gradually, through successive runs of the device through a head of hair, emptied from back to front. In practice, each of said individual runs will begin as the squeeze plate **44** and body plate **50***a* appear in the position depicted in FIG. **24**A and will gradually follow, through successive individual runs, the entire FIG. **24** mechanical sequence until the squeeze plate **44** and body plate **50***a* meet face to face as depicted in FIG. **24**D having emptied the color container **51***j*.

[0195] This back to front emptying process of the color container **51***j* is assured as the dual elbow spring **48***b* is located at, and therefore, creates tension between the front of the squeeze plate **44** and the front of the body plate **50***a*, thereby, tensioning the front of each of the two plates away from one another to the open most position. This tension is maintained as one elbow of the dual elbow spring is attached on one side to the body plate **50***a* and on the other side to the squeeze plate slide hinge pin **49***b*. The attachment between the spring and the front of the squeeze plate is maintained as the bottom squeeze plate slide hinge pin **49***b* extends through the center of one of the two spring **48***b* coils. This spring **48***b* coil attachment is also the pivot point of a second preferred squeeze plate **44** tension. This front pivotal tension urges the back of the squeeze plate **44** to the open most position away from the back of the body plate **50***a* when the trigger **29***a* is released. Since both of the outward tensions described above are located at the front of the two plates, inward pressure applied to the middle of the squeeze plate **44** by the contact point **29***b* of the trigger **29***a* will, through successive runs of the device through a head of hair, cause the back of the squeeze plate **44** to move toward and contact the back of the body plate **50***a* first, followed by the approach of the front of the squeeze plate **44** toward the front of the body plate **50***a*.

[0196] As seen in FIGS. **23**B and **24**A, the point in each individual run where the trigger **29***a* is released causes the trigger **29***a* to pivot out to the open position with the tension of the trigger spring **32**. Said mechanical action causes the squeeze plate **44** to move away from the body plate

50*a*, thereby, returning the squeeze plate to the open resting position against the tension of the dual elbow squeeze plate spring **48***b*. Concurrently, the trigger slide **35** returns to the resting position and, in doing so, slides off of the slide wedge contact angle **40***b*. This release of the slide wedge contact angle **40***b* causes the slide wedge **40***a* to slide upward with the tension of the slide wedge spring **40***d*, which, in turn, raises the wedge section **40***c* of the slide wedge **40***a* out from between the rack slide **42***a* and the side of the gear box **6**. The rack slide **42***a* is then released to slide inward, guided by the rack slide seat **43** in which it is slidably seated, toward the side of the gear box **6** with the tension of the rack slide spring **42***b*. Finally, as one side of the rack gear **7** is attached to the rack gear slide **42**, the inward motion of the rack gear slide **42***a* causes the rack gear **7** to move back to the resting position along with the pinion gear **8** and therefore the hook **2***a*.

[0197] The color container valve, neck and coupling are the same as the color containers described in the multi-hooking applicator embodiment described earlier in this disclosure only embodied in the singular.

[0198] As the device has been described in use above, the hooks move from the open position FIG. **1**A to the closed position FIG. **1**B, thereby, entraining a section of hair into the closed position FIG. **1**B. As this occurs, each entrained section of hair becomes automatically positioned directly under the color aperture **5** of each applicator nozzle **11***a*. Now, as the operator squeezes the handle of the device and draws the device away from the scalp, the color exits each aperture **5** and deposits a bead of color only onto the top of each entrained section **1***b*. The entrained sections **1***b*, therefore, exit the closed position with a bead of color **1***d* applied to the top of each entrained section **1***b*. This is because (as seen in FIG. **3**D) the distance between the hair color aperture **5** of the applicator nozzle **11***a* and the front foot channel **4***c* is quite small, causing the entrained sections **1***b* to exit the closed position before the beads of color **1***d* have an opportunity to be thoroughly pressed into the entrained sections of hair **1***b*. Although the controlled depositing of a bead color **1***d* onto entrained sections of hair **1***b* is a useful and novel feature of the device, a more complete saturation of the hair color **1***d* into the entrained sections of hair **1***b* may be desired.

[0199] An effective way for the beads of color **1***d* to be suffused completely into the bundles of hair **1***b* as they convey through the device is to elongate the closed position. An elongated closed position gives the color, as it exits the color aperture **5** of the applicator nozzle **11***a*, more time and space to fully press into and saturate the entrained section **1***b* before the color treated entrained section **1***d* exits the front of the closed position.

[0200] It may be assumed that simply widening the hook **2***a* and applicator nozzle **11***a* would sufficiently serve this purpose; however, a limitation becomes apparent when the hook **2***a* is widened. When the hook **2***a* is widened slightly beyond what has been presently depicted, the hook **2***a* becomes too wide to effectively entrain a section of hair **1***b*. The following embodiment includes the additional structures that allow the closed position to be elongated without widening the initial hair entrainment portion of the hook **2***a*.

[0201] As Seen in FIGS. **25**A and **25**B, the hooking applicator **1***a* is shown in the same open and closed position respectively as seen in FIGS. **1**A and **1**B. Notice, also, in FIGS. **25**A and **25**B the additional structural features attached to and extending forward from the fronts of both the hook **2***a* and front foot **4***a*. The structure attached to the front of the hook **2***a* is the hook extension **100***a*; the triangular, visco-elastic foam filled structure attached to the front foot **4***a* is the foot extension **101***a*. When the hook extension **100***a* and the foot extension **101***a* are in the closed position relative to one another (as seen in FIG. **25**B) with a section of hair **1***b* entrained between, the hook extension **110***a* and foot extension **101***a* form the saturation chamber **102***a*. This saturation chamber **102***a* represents a second stage of color application. The hook and applicator in the closed position is therefore the first stage. With the addition of the hook extension **100***a* and foot extensions **101***a* (second stage), the liquid color has an additional foam channeled distance to travel through, along with the entrained section of hair, thereby, giving the liquid hair color more opportunity to blend into the entrained section.

[0202] FIGS. **26**A through **26**D depict a series of bottom views of the hooking applicator **1**a. These figures show the process of hair entrainment from the fully open position, through intermediate rotational hook **2**a positions, to the fully closed position with a section of hair **2**b entrained within.
[0203] Notice as the hook **2**a and hook extension **100**a move from the fully open position depicted in FIG. **26**A to the intermediate position depicted in FIG. **26**B, a section of hair **1**b has been entrained by just the hook **2**a. The hook extension **100**a, at this point has not yet interacted with the entrained section of hair **1**b.
[0204] FIG. **26**C depicts the next rotational position in series and shows the entrained section **1**b having been entrained by the hook **2**a and making initial contact with the hook extension **100**a. As depicted, the hook extension **100**a is attached to the front of the hook **2**a a distance away from the point of the hook **3**a. This arrangement allows the hook **2**a to fully entrain the section of hair **1**b prior to encountering the hook extension **100**a. This represents a preferred method of widening the closed position **2**a without jeopardizing the ability of the hook **2**a to consistently entrain a section of hair **1**b.
[0205] FIG. **26**D depicts the hooking applicator **1**a along with the extensions **100**a, **101**a thereof having entrained a section of hair **1**b into the fully closed position. Notice the entrained section of hair **1**b is fully positioned within the entire width of the combined hook **2**a and hook extension **100**a. Also, notice that the hook extension **100**a is attached to the hook **2**a in such a way that the inside surface of the hook extension **110**a is attached to the outside surface of the hook **2**a (see FIGS. **26**A-**26**D and FIG. **27**A for front view of this placement). This placement provides an advantage. As the hook **2**a rotates and entrains a section of hair **1**b, this lower position of the hook extension **100**a relative to the hook **2**a combined with the slightly rounded front corner **100**b of the hook extension **100**a allows it to slide easily under the entrained section **1**b rather than bumping into it and pushing it to the side.
[0206] Now that the present embodiment featuring the saturation chamber has been described in the context of hair entrainment, what follows is a description of said embodiment in terms of controlled and thorough color application onto the entrained section of hair.
[0207] While viewing FIGS. **27**A and **27**B Notice the foam seal **101**b occupying the space within the foot extension **101**a. A section of foam **101**b is folded and the corner of the fold is attached to the inside corner **101**c of the foot extension **101**a with both sides of the fold extending out; the distal end of each side of the folded section wraps around the outside edges **101**d of the foot extension **101**a and are attached to each respective outside edge **101**d.
[0208] While viewing FIGS. **27**A and **27**B in series notice, as the hook **2**a pivots from the open to the closed position, the outside edges **100**c of the hook extension **100**a come into contact with and firmly compress the foam wrapped edges **101**d of the foot extension **101**a. Also, while in the closed position **27**B, the foam is thick enough that most of the inside surface of the hook extension comes into light compression contact with the foam from the outside edges toward the fold, with the fold being the narrow channel that does not come into contact with the hook extension. This foam arrangement creates a seal along both foam wrapped edges **101**d of the foot extension **101**a thereby preventing color from leaking from these areas, while the fold of the foam (foam channel **101**e) directs the color, along with the entrained section of hair through, and then out of the front of, the closed position.
[0209] This channeling together of the hair section and hair color through this additional length of closed position **102**a assists in causing a fully color saturated section of hair to exit from the front of the saturation chamber **102**a. The thickness and density of the foam may be adjusted in order increase or decrees the size of the foam channel and therefore, the flow of color or to accommodate differences in color viscosity.
[0210] Varying, in particular ways, the inside geometry of the closed position also assures that a fully color saturated section of hair emerges from the closed position. The lower position of the hook extension **100**a described above creates such a variation of said inside geometry. This

preferred variation is in the form of a step down **100***d* from the front edge of the hook **2***a* to the back edge of the hook extension **100***a* (see FIG. **26**A-D). This step down **100***d* creates an open space under the entrained section of hair **1***b* as the entrained section **1***b* is still in the closed position. Now, as the hair color flows out of the aperture **5**, the color flows onto the top of the entrained section **1***b* in the first stage of color application, then as the color flows from the first stage to the second stage of color application, the space beneath the entrained section **1***b* at this step down **100***d* point gives the color an opportunity to flow under the entrained section of hair **1***b* while the entrained section of hair **1***b*, moves through the closed position. So the additional foam channeled distance of the closed position travelled in combination with this step down **100***d* point along the way causes the entrained section **1***b* of hair to become entirely coated with hair color before exiting the closed position.

[0211] The following describes an embodiment of the invention in the form of another modified hair comb type grooming appliance. Said comb is mechanically modified to entrain sections of hair into the closed position as well as apply hair color to said hair sections.

[0212] In order for the comb to perform the first of these functions, a series of hooks move from the open to closed position between some or all of the teeth of the comb. The preferred embodiment depicted in FIG. **28**A thru **28**C show 4 of said hooks **103***b* resting in the open position in front of every other tooth **104***a* for a total of 4 comb teeth **104***a* with a hook **103***a* in front. Notice each hook **103***a* is attached to the slide member **103***b*. This slide member **103***b* extends across the spine **104***b* of the comb section **104***c* and continues to extend back across the top of the handle **105***a*. The slide member distal end **103***b* is angled down toward the bottom of the handle hinge **105***b* with the slide member distal end **103***b* touching the contact point seat **105***d* of the hinge bifurcation **105***c*. Now, when an operator squeezes the handle **105***a*, the handle hinge **105***b* pivots forward causing the contact point seat **105***d* of the hinge bifurcation **105***c* to push forward on the slide member distal end **103***b*. As the hooks **103***a* are attached to the slide member **103***b*, the hooks **103***a* therefore move from the open to closed position by squeezing the handle **105***a*. Like the squeeze handle **105***a* action of previously described embodiments, the first increment of squeeze of the present embodiment (see FIG. **28**B) causes the hooks **103***a* to move from the open to closed position while the second increment of squeeze (see FIG. **28**C) applies color dispensing pressure to the color container **106***a* while maintaining the hooks **103***a* in the closed position. The bifurcation spring **105***e* of the hinge bifurcation **105***c* allows the hooks **103***a* to advance no farther than the closed position while the handle **105***a* may continue to close to an ever greater extent to the point where the color container **106***a* is empty. The handle hinge **105***b* and bifurcation **105***c* move in unison until the hooks **103***a* reach the closed position. Once the hooks **103***a* are closed, the hinge **105***b* will continue to pivot forward but the bifurcation **105***c* will remain stationary while maintaining forward pressure on the slide member **103***b* and therefore the hooks **103***a* in the closed position. This is because the bifurcation spring **105***e* does not have enough tension to resist the pivot of the handle hinge **105***b* yet has enough tension to maintain pressure on the closed position of the hooks **103***a*. When the operator releases the handle **105***a*, the hooks **103***a* return to the resting position. This occurs as the slide member distal end **103***c* contiguously rests within the contact point seat **105***d*. As the operator squeezes the handle **105***a* the contact point seat **105***d* pushes the slide member **103***b* forward; conversely, as the operator releases the handle **105***a* the contact point seat **105***d* pulls the slide member **103***b* back.

[0213] In use, the operator draws a preferably 4 to 10 cm straight parting of hair along a scalp of hair using the parting stem **107**. The operator then gathers together the hair comprising the top of the parting and secures it temporarily up and away from the parting exposing the part line **1***c* along the scalp. Next, the operator draws another parting under, adjacent to and parallel with the first parting and maintains the parting stem **107** in this position parallel to the scalp and the first parting with the section of hair draped over it. The distance between the first and second parting is preferably between 1 to 10 mm. The operator will now carefully move the parting stem **107** away from the scalp causing the draped section of hair to separate out and away from the rest of the hair. With the section of hair **1***h* now separated away from the scalp **1***f*, the operator, with her free hand, will lift the draped section of hair **1***h* away from the parting stem **107** and hold it there. She will pull the section of hair **1***h* comfortably taut so that all of the hair in the section **1***h* is projecting straight away from the scalp **1***f* at a preferably 90 degree angle. The operator will now turn the device so that the comb section **104***c* is flatly facing the scalp **1***f*, and is preferably above the separated section of hair **1***h* with the teeth **104***a* of the comb **104***c* pointed down toward the section of hair **1***h*. She will push the teeth **104***a* into the section of hair **1***h* so that the entire section separates into individual bundles of hair **1***g* between all of the teeth **104***a* of the comb **104***c*. She will continue to push the teeth **104***a* of the comb **104***c* into the section of hair **1***h* until each section of hair **1***g* between each of the teeth comes into contact with the spine **104***b* of the device. See FIG. **29**A depicting the bundles of hair **1***g* separated between the teeth **104***a* against the spine **104***b*. The operator will, at this point, begin to squeeze the handle **105***a*.

[0214] See FIGS. **28**A and **28**B in series for the following. As the operator squeezes the handle **105***a*, the handle hinge **105***b* pivots forward causing the contact point seat **105***d* of the handle hinge bifurcation **105***c* to push forward on the slide member distal end **103***c* and therefore the slide member **103***b*. This causes each hook **103***a* to slide from the open position to the closed position by crossing each respective space between the comb teeth **104***a*. See FIGS. **29**A and **29**B in series for the following. As each hook **103***a* travels across each space, each hook **103***a* will encounter a stalk of hair **1***g* positioned within each space. In this manner, each hook **103***a* entrains a bundle of hair **1***b* and pushes the bundle of hair **1***b* against the applicator **108***a*.

[0215] The operator has, at this point, entrained bundles of hair **1***b* between every other comb tooth **104***a*. This, of course, means that there is a stalk of hair **1***g* between every other comb tooth **104***a* that is not entrained. As the operator has pushed the teeth **104***a* of the comb **104***c* down onto the projected section of hair **1***h*, the operator can simply let go of the projected section of hair **1***h* with the hand that is not holding the device and the hair **1***g* that is not entrained into the closed position will simply fall out from between the teeth **104***a* of the comb **104***c* and into the rest of the hair **1***f*.

[0216] Liquid hair color is moved from the color container **106***a* to the applicators **108***a* through a tubular channel formed lengthwise along the neck **104***d* and spine **104***b* forming a primary supply line **109***a* (see cross section view FIG. **30**). As viewed in FIG. **31**, this supply line **109***a* runs along the opposite side of the comb spine **104***b* from the hooks **103***a* and applicators **108***a*. As depicted, this primary supply line **109***a* branches toward and opens out of the bottom of every other tooth **104***a* into each color applicator **108***a*; therefore, squeezing the handle **105***a* loaded with a container **106***a* of hair color causes the hair color to exit from each color applicator **108***a* and onto the entrained sections of hair **1***b*.

[0217] The applicators **108***a* are very similar in function to the applicators described earlier. As seen in FIG. **32**, each applicator housing **108***b* is preferably composed of a rectangular box that is open on the side facing the hook hollow **103***d*. The applicator housing **108***b* is preferably occupied by an insert of viscoelastic foam **108***c* that partially extends out of the applicator housing **108***b* toward the direction of the hook hollow **103***b*. The front of this outwardly extending portion of the foam insert is the applicator face **108***d*. As the applicators **108***a* are positioned along the front of the comb section **104***c*, a channel **108***e* is formed into the foam face **108***d* extending from the front of the applicator **108***a* to the opening in the foam **108***f*. As seen in FIG. **31**, a branch **109***b* from the supply line **109***a* opens into the applicator housing (see FIG. **32**) and this opening connects to the foam opening **108***f*. This face channel **108***e* is in line with the section of hair **1***g* that occupies the space between the teeth **104***a* once the hair section **1***g* is brought into contact with the spine **104***b* of the comb (see FIGS. **29**A-**29**B). When the hook **103***a* closes over the section of hair **1***b* as such, the section of hair **1***b* will be pressed into this face channel **108***e* as well as the opening in the foam **108***f*. With the section of hair **1***b* entrained into the closed position, liquid hair color will travel through the supply line **109***a* and into the applicator housing **108***b*. From this point the color continues to move through the opening in the foam **108***f*, onto the entrained section of hair **1***b* and through the face channel **108***e* where the color then exits the closed position only from the front along with the color coated entrained section of hair **1***b* (see FIGS. **29**, **30**, **31**, **32**).

[0218] FIGS. **33**A-B depict an additional embodiment of the foot extension **101***a*′. This embodiment is similar in shape and function to the foot extension **101***a* described in FIGS. **25**A-B. An additional feature of the foot extension **101***a*′ seen in FIGS. **33**A-B is the rubber facing **101***f* attached to the front. FIG. **33**B depicts an entrained section of hair **1***b* emerging from the closed position as would be seen immediately as the operator begins to squeeze the handle and draw the device away from the head of the recipient. As the entrained section of hair **1***b* emerges from the closed position, it does so through a squeegee type tension between the top surface of the hook extension **100***a* and the bottom of the rubber facing **101***f*. This occurs as the rubber facing **101***f* is fixed flat to the front edge of the hollow foot extension **101***a*′. As the hook extension **100***a* is in the closed position against the foot extension **101***a*′, this facing **101***f* is open only along the bottom edge that contacts the hook extension **100***a*, between the inside edges of the foot extension **101***a*′. Also, this opening is located over the area of the hook extension **100***a* that the entrained section **1***b* passes. This arrangement precisely channels the color onto the entrained section **1***b* and causes the color to dispense more uniformly. The embodiment depicted in FIG. **25**A-B may also utilize the rubber facing **101***f*.

[0219] FIGS. **34**A-B depicts the hooking applicator **1***a* with the present intention of featuring the additional embodiment of the foot extension **101***a*′, with the rubber facing **101***f* removed from the front; this, in order to more entirely view the nozzle seal **11***b*, and foot extension seal **101***b*, as each are depicted in halftone (The nozzle seal **11***b* and foot extension seal **101***b* are actually one continuous seal). Notice, every surface joining the hook **2***a* and hook extension **100***a* with the applicator nozzle **11***a* and foot extension **101***a* in the closed position is sealed in said closed position.

[0220] Notice the difference between the foot extension seal **101***b* in FIGS. **27**A-B and FIGS. **34**A-B. Said FIGS. **27**A-B seal nearly fill the saturation chamber **102***a* thereby utilizing the shape of the seal fold or foam channel **101***e* to direct the color exclusively onto the entrained section of hair as the entrained section passes through and emerges from the saturation chamber **102***a*. In comparison, the FIGS. **34**A-B seal **101***b* occupies only the surfaces of the foot extension **101***a* that come into contact with the hook extension **100***a* while leaving the space within the saturation chamber **102***a* open. Therefore, the sole purpose of the FIGS. **34**A-B seal **101***b* is to prevent color from leaking from the closed position and allowing color to flow exclusively from between the center bottom edge of the rubber facing (see FIG. **33**B, **101***f*) and the front, top of the hook extension **101***a*. This seal **101***b* also allows the inside of the saturation chamber **102***a* to fill with color, helping to saturate the entrained section **1***b* before emerging from the saturation chamber **102***a*.

[0221] As the space within the saturation chamber **102***a* is open in FIG. **34**A-B, any number of color spreading elements may be incorporated into the space. In addition to the step down **100***d* described earlier in FIGS. **26**A-D, these elements may include a ribbed **100***f* inside surface of the hook extension **100***a* (see FIG. **35**). Bristles **101***h* may extend down from the inside of the foot extension **101***a* (see FIG. **36**). Also, an internal squeegee **101***i* may be positioned inside the foot extension **101***a* (see FIG. **37**). Numerous geometric variations of the inside of the closed position will interact with the entrained section of hair and divert the color in and around the entrained section as it moves through the closed position.

[0222] The following description will be referring to surfaces in straight front views FIGS. **38**A-D. FIGS. **38**A-D. As these are not perspective views, the precise location of several of the referenced surfaces are not visible and are actually located on the opposite side of the surface on which the lead line points. This inconvenience is necessary as the primary interaction of surfaces described is best depicted in this straight front view. For a more precise location of the surfaces described in this section, refer also to FIG. **34**A-B Continuing with the differences between the two embodiments, notice that the foot bridge **4***d* as well as the portion of the foot extension **101***a* extending up from the foot bridge **4***d* is narrower and positioned farther toward the outside right on the FIGS. **38**A-D embodiment than that of the like surfaces of FIGS. **27**A-B embodiment. Positioning the foot bridge **4***d* in this manner allows the hook point **3***a* to pivot farther away from the scalp **1***f* and closer to the closed position before encountering the foot extension seal scissors edge **101***g*. See FIGS. **38**A-D in sequence and notice the position of the hook point **3***a* as it pivots from the open to the closed position. FIGS. **38**A and **38**B depict the hook point **3***a* pivoting toward and along the scalp **1***f* respectively, while FIGS. **38**C and **38**D, having entrained a section of hair **1***b*, is pivoting away from the scalp **1***f* and has pivoted farther before encountering the foot extension seal scissors edge **101***g*, as compared to the FIGS. **27**A-B embodiment. This increased distancing factor allows the entrained section of hair **1***b* more opportunity to slide farther back on the hook hollow **2***d* before the hook point **3***a* encounters the foot extension seal scissors edge **101***g*. This distancing factor also prevents the hook point **3***a* from continuing to entrain hair **1***b* too close to the position that the hook point **3***a* is encountering the foot extension seal scissors edge **101***g*; this creates an opportunity for strands of hair to become pinched between the hook **2***a* and said seal edge **101***g* while in use. In like manner, this hair pinching issue is also avoided in the context of the entrained section **1***b* being consequently scooped into the hook extension **100***a*; the entrained section of hair **1***b* has more opportunity to slide farther back on the hook extension **100***a* prior to the hook extension contact edge **100***e* making contact with foot extension seal **101***b.*

[0223] Described here is an alternative mechanical means for coordinating, in series, the opening and closing of the hooks with the dispensation of the hair color by a squeezing and releasing of the handle of the device.

[0224] An explanation of how the lever **24***a* interacts with the lever button **24***d* in order for the lever **24***a* to act on the level pallet **22** as a mechanical means of squeezing hair color out of the color container **51***a* as the handle **20***a′* is squeezed has been described previously in this disclosure (See description of FIGS. **10**A-D). This mechanical operation is functionally identical to this earlier description, and therefore, will not be described in the following. What is different is the mechanical means responsible for coordinating the squeeze and release of the handle **20***a′* with the opening and closing of the hooks **2***a*. See FIGS. **39**A-D for a description of these mechanical means.

[0225] The FIG. **8** drawing depicts a single rod seemingly bent at multiple right angles in order to form the head mounts **15** and attached to the top handle section **20***b* of the handle **20***a* as a separate part. Notice that the head mounts **15′** in FIG. **39**A are flat, wide and are molded in such a manner that the top handle section **20***b′* and the head mounts **15′** are one continuous part. The present FIG. **39**A version of the head mounts **15′** is more conducive to the rack slide assembly that will be described in the following as well as being an improvement from a manufacturing and aesthetic perspective.

[0226] FIG. **39**A also depicts the head **14***a* apart from the head mounts **15′** in order to show how the head **14***a* is attached to this alternative head mount **15′** of the FIG. **39**A embodiment, and to more clearly show the head bracket **14***b* and rack gear **7***a* unobstructed by the parts that appear in front of it.

[0227] Continuing to view FIG. **39**A, with the head **14***a* assembled to the head mounts **15′**, notice that the ends of the head brackets **14***b* as well as the end of the rack gear **7***a* are seated within, extending out of but not fixed to the head bracket seats **14***g* and the rack gear seat **14***h*. The head brackets **14***b* and rack gear **7***a* end on the side of the head **14***a* that is not visible is also seated in like manner but not fixed. This allows the head **14***a* to more easily flex into the curve of the scalp. A removable peg **14***i*, cotter pin or the like may also be seated at one or more distal ends of the head bracket **14***b* to allow freedom of movement of the head brackets **14***b* as they are seated in the head bracket seats **14***g* while eliminating the possibility of an unintentional disengagement of the head brackets **14***b* from the head mounts **15**′.

[0228] The following describes the series of mechanical motions that causes the initial squeezing of the handle **20***a*′ to bring the hooks **2***a* to the closed position. This series of mechanical motions involves the rack gear **7***a*, split slide **110**, split slide actuator **111***a*, actuator tine **112***a* and pivot rod **113***a*; these parts will be referred together as the rack slide assembly.

[0229] See FIG. **39**A-D for the following description of the functional sequence of the rack slide assembly. FIG. **39**B includes a transparent head mount **15**′ and a partial exploded view of the rack slide assembly in order to provide greater visual clarity throughout the entire rack slide assembly description.

[0230] FIG. **39**A-B shows the device and therefore the rack slide assembly in the open resting position.

[0231] FIG. **39**C shows the handle **20***a*′ having been compressed from the fully open position to the handle **20***a*′ position that fully engages the hooks **2***a* without pressing against the color container **51***a*. Also, notice that this partial compression of the handle **20***a* causes the tine tip **112***b* of the actuator tine **112***a* to move up the split slide actuator **111***a*. This upward movement of the actuator tine tip **112***b* against the split slide actuator **111***a* causes the split slide actuator **111***a* to pivot at the head mount pivot point **111***b*. As the split slide actuator **111***a* pivots on the head mount pivot point **111***b*, the top of said actuator **111***a* levers forward while the bottom of said actuator **111***a* levers back. Consequently, the split slide **110***a* moves back as the split slide peg **110***c*, which is located at the rear of the split slide **110***a*, is seated to the bottom of the split slide actuator **111***a* at the split slide actuator eyelet **111***c* (also see FIG. **39**B for actuator eyelet **111***c*).

[0232] Now, while the split slide **110***a* moves back, the wedge contact point **113***b* of the pivot rod **113***a* slides along the gradually outward extending angle of each wedge section **110***b* of the split slide **110***a* causing the wedge contact point **113***b* of the pivot rod **113***a* to move to the side. The pivot rod **113***a* is strictly confined to a side pivot as the opposite end of the pivot rod **113***a* is positioned within the pivot rod seat **113***c* (see FIG. **39**A for pivot rod seat **113***c*).

[0233] (Still viewing FIG. **39**C) notice, as the wedge contact point **113***b* of the pivot rod **113***a* is positioned within the pivot rod eyelet **113***d* of the rack gear **7***a*, the side movement of the pivot rod **113***a* causes the rack gear **7***a* to likewise move to the side (see FIGS. **39**A-B for pivot rod eyelet **113***d* and rack gear **7***a*). The split configuration of the split slide **110***a* positions one wedge section **110***b* of the split slide **110***a* above the rack gear **7***a* and one wedge section **110***b* below the rack gear **7***a*. This split configuration allows a balanced sideways tension to be exerted on the rack gear **7***a* by the wedge contact point **113***b* of the pivot rod **113***a*. The distance that the rack gear **7***a* moves is equal to the width of the wedge section **110***b* of the split slide **110***a*, therefore, the distance that the hooks **2***a* pivot is determined by the same.

[0234] Finally, as seen in FIG. **39**C, the rack gear **7***a*, having been moved to the side by the wedge contact point **113***b* of the pivot rod **113***a*, causes the hooks **2***a* to pivot to the closed position.

[0235] FIG. **39**D shows the split slide actuator **111***a* remaining in the position depicted in FIG. **39**C even as the handle **20***a* is in the fully closed position with the tine tip **112***b* of the actuator tine **112***a* lifted out of contact with the split slide actuator **111***a*. This allows the handle **20***a* to be fully squeezed to the closed position while maintaining the hooks **2***a* in the closed position.

[0236] When the handle **20***a*′ is released, the tine tip **112***b* moves down and encounters the split slide actuator stop **111***d*. Said stop **111***d* is the widened, sloped portion at the bottom of the split slide actuator **111***a*. When the tine tip **112***b* pushes down on the split slide actuator stop **111***d*, with the tension of the handle hinge spring **21***b*, the split slide actuator **111***a* pivots back to the resting position depicted in FIG. **39**A-B, along with the remaining parts of the rack slide assembly, thereby causing the rack gear **7***a* and hooks **2***a* to return to the open position.

[0237] The following is a description of a useful embodiment feature that enables the operator of the device to adjust the size of the entrained sections of hair. While viewing FIGS. **40**A-B notice the dial **114***a* positioned on the right head mount **15**′. The purpose of this dial **114***a* is to adjust the distance between the tips of the hooks **2***a* and the inside edge of each foot bridge **4***d*. The space between said parts (generally within the range of: the point where the hook tip **3***a* has pivoted close enough to the scalp for hair entrainment to occur, and, the point where the hook tip **3***a* encounters the foot bridge **4***d*) is approximate to the width of the section of hair that is entrained by each hook **2***a*. This dial **114***a* consists of a small raised circular disc **114***a* that is turned by the finger and thumb tip. A shaft **114***b* extends fixed and centered on the inside of this dial **114***a*, through an opening in the right head mount **15′**. This opening is approximately the diameter of the shaft **114***b*. The shaft **114***b* extends out of the opposite side of the head mount **15′** and is fixed to another oval shaped disc **114***c*, therefore, turning the outside disc **114***a* causes the inside disc **114***c* to turn. With the handle in the fully open position, the bottom edge of the split slide actuator tine **112***a* is in contact with the top edge of the oval disc **114***c*. In this manner, the oval disc **114***c* functions as the rest stop of the bottom handle section. Now, with the handle released to the open position, the operator may turn the dial **114***a* back and forth one half of a complete rotation. As this occurs, it will be noticed that the two hinge sections that make up the handle will slightly close toward and open away from one another in tandem with the turning of the dial **114***a*. This slight opening and closing occurs as the oval disc **114***c* attached to the dial **114***a* is, again, an oval **114***c* and therefore, causes the top edge of the oval disc **114***c* to be higher or lower relative to the shaft **114***b* and dial **114***a*. See FIG. **40**A to view the dial **114***a* in the low position and FIG. **40**B to view the dial **114***a* in the high position.

[0238] As described earlier, the rack slide assembly is engaged and therefore the hooks **2***a* are turned by the initial squeezing of the handle. In like manner, the slight initial squeezing of the handle that is caused by the turning of the dial **114***a* also causes the hooks **2***a* to turn slightly. Consequently, different dial **114***a* positions cause the space between the hook tips **3***a* and the foot bridges **4***d* to vary. FIG. **40**A shows the dial in a position that causes the space between the hook tip **3***a* and foot bridge **4***d* to be wider while FIG. **40**B shows the dial **114***a* in a position that causes the space between the hook tip **3***a* and foot bridge **4***d* to be narrower.

[0239] This dial **114***a* is attached to the head mount **15′** tightly in order to create enough turning friction that the dial **114***a* remains in place once the dial **114***a* is set and released by the operator. An arrow or dot may be placed on the edge of the dial **114***a*. This arrow or dot may line up with indicator markings place on the head mount **15′**. These markings partially surrounding the dial **114***a* so that, as the dial **114***a* is turned, the dot or arrow may line up with the markings as an indication of specific widths of entrained hair bundles.

[0240] FIG. **41** depicts a preferred embodiment of the device including a head hood **115***a* and a hinge hood **116**. The head hood **115***a* and hinge hood **116** represent an aesthetic as well as an ergonomic and safety feature. The head hood **115***a* is hinged **115***d* to the front of the top section of the handle **20***b*. The head hood **115***a* is held down in place by the hood latch **115***b* and hood peg **115***c*. As depicted in FIG. **41**A, the head hood **115***a* may be lifted by turning the latch **115***b* up and away from the peg **115***c* and simply lifting the head hood **115***a*. Both the head hood **115***a* and the latch **115***b* may be spring loaded so that the head hood **115***a* lifts open automatically when the latch **115***b* is lifted. The front of the latch **115***b* is shaped and positioned in order that when the head hood **115***a* is closed, the side of the peg slides along the front of and lifts the latch. The once the hood is fully closed, the spring loaded latch **115***b* automatically engages with the peg **115***c*.

[0241] Also depicted in FIGS. **42**A and **42**B is the manifold retractor **117***a*. When the operator loads a color container **51***a* into the front of the device, the operator will lift the head hood **115***a* thereby exposing the manifold **53***a*. As the manifold hoses **53***b* are composed of rubber, the operator may grab the manifold retractor **117***a* on the front of the manifold **53***a* and bend the manifold **53***a* down and place the manifold retractor **117***a* onto the manifold hook **117***b* thereby holding the manifold **53***a* down and out of the way in order to allow the operator better access to load a color container **51***a* into the front of the device.

[0242] The following (See FIG. **43**A-C) is a description of yet another Hooking Applicator **1***a* iteration utilizing the squeegee action between the bottom edge of a rubber or flexible facing **101***f* (or, from this point forward, what will be called the 'squeegee' **101***f*') and the top surface of the hook extension **100***a* (see FIG. **33**A-B for previous hook extension **100***a*) in a more controlled manner; and this, while integrating the structure and action of the hook and the extension **100***a* more completely into the main body of the hook **2***a*'. Now, this hook extension **100***a* (or what will now be referred as the hook offset **118***a*) occupies the entire front edge of the hook **2***a*' and also extends to the hook point **3***a*'. This now wider hook **2***a*' allows the sections of hair to be entrained onto the hook offset **118***a* more easily than the previous hook extension **100***a*, as the hooked hair would often get caught on the side edge of the hook extension **100***a* and therefore fail to occupy the effective top surface of the hook extension **100***a*. Essentially, the entire hook **2***a*' became wider with a hook offset **118***a* along the front edge, and the extension **100***a* was eliminated.

[0243] Another hook feature presently described is the hook shield **118***b*. This hook shield **118***b* is attached to the front edge of the offset **100***a*' in such a manner that the back surface of said hook shield **118***b* is contiguous to a portion of the bottom front surface of the squeegee **101***f*', as the hook **2***a*' and applicator nozzle **11***a* are in the closed position. Now, with the hook **2***a*' and applicator nozzle**11***a* in the closed position, a portion of the bottom of the squeegee **101***f*' is sandwiched between the back surface of the hook shield **118***b* and the front edge of the nozzle contact surface **118***c* of the hook **2***a*' with the entire length of the bottom edge of the squeegee **101***f*' in contact with the top surface of the hook offset **118***a*. Also, with the hook and applicator in the closed position, the open portion of the front and back of the squeegee **101***f*' (that is, the portion of the bottom front and back surfaces of the squeegee **101***f*' that is not sandwiched between the front edge of the nozzle contact surface **118***c* of the hook **2***a*' and the back surface of the hook shield **118***b*) is located along the front of the color channel **2***b*. This arrangement seals the sandwiched portion of the bottom of the squeegee **101***f*' while confining an un-sandwiched yet bottom sealed section of the bottom of the squeegee **101***f*' to the front of the color channel **2***b*. Another feature of the squeegee **101***f*' that adds yet another element of color distribution control to the action of the bottom edge of the squeegee **101***f*' is the squeegee recess **101***h*. This squeegee recess occupies the portion of the front side and bottom edge of the squeegee that is positioned in front of the color channel **2***b* of the hook **2***a*'. This squeegee recess **101***h* is thinner than the remaining portion of the squeegee **101***f* as said squeegee recess **101***h* is the portion of the squeegee **101***f*' that conforms to the entrained section of hair and allows hair color to coat the entrained section as said entrained section conveys through the closed position. If said squeegee recess **101***h* portion of the squeegee **101***f*' were as thick as the rest of the squeegee **101***f*', the bottom edge of the squeegee **101***f*' would pinch the entrained section of hair against the hook offset **118***a* with more force than is functional. In addition to said malfunction, the liquid hair color would be prevented from coating the entrained section of hair as said hair conveys through the closed position. However, this thicker portion of the squeegee **101***f*' allows for a stronger seal for the locations along the closed position that are meant to seal while in said closed position.

[0244] Now, as the hook **2***a*' pivots and entrains a section of hair within the closed position between the hook **2***a*' and the applicator nozzle **11***a*, the side edge of the hook shield **118***b* (the side edge adjacent to the color channel **2***b*) gathers the entrained hair directly over of the color channel **2***b*. Now, the entrained hair is gathered directly over the color channel **2***b* from both sides, one side from the hook shield **118***b* and the other side from the scissors edge **4***f* of the foot **4***a*. Now that the entrained section of hair has been gathered over the color channel **2***b* of the hook **1***a*', the entrained section is also sandwiched between the hook offset **118***a* and the bottom edge of the squeegee recess **101***h*. Now, as the hair moves through the closed position, and as color is dispensing onto the entrained section of hair, the color is spread or 'squeegeed' onto the entrained hair in a controlled manner. This control stems from the visco-elastic property of the squeegee **101***f*' as the bottom edge of the squeegee recess **101***h* of the squeegee **101***f*' will conform around the entrained section of hair through a range of densities of entrained sections, while maintaining the seal against the hook

offset **118***a* on either side of the entrained section. In addition to this, as described above, the squeegee recess **101***h* in front of the color channel **2***b* is not sealed front to back as is the rest of the bottom of the squeegee **101***f*′. This allows for the full ‘squeegee action’ of the squeegee **101***f*′ as the squeegee recess **101***h* can flex forward slightly (as per said ‘squeegee action’) as the entrained section of hair moves through the closed position, thereby allowing the liquid color to coat the entrained section of hair as it slides between the squeegee recess **101***h* and the hook offset **118***a* without said hair color leaking from any other point along the bottom of the squeegee**101***f*′.
[0245] Another benefit of this squeegee **101***f*′ arrangement is the fact that, although the squeegee recess **101***h* is free to move forward when acted upon by the movement of the hair through the closed position, and thereby allow liquid hair color to escape onto the entrained section as the section moves through the closed position; without the presence of hair moving through this closed position, the bottom edge of the squeegee recess **101***h* in front of the color channel **2***b* will remain sealed to the viscous hair color even under moderate liquid pressure inside the closed position. This allows for several liquid control factors:
[0246] The flow of the liquid from the color container **51***a* into the (hair section occupied) closed position is aerated, as air can escape under the internal pressure caused inside of the closed position (by the squeegee **101***f*′ and the nozzle seals **11***b*) faster than the viscous hair color can dispense. This prevents dry spots from occurring on the color treated sections of hair, during a pass of the device, from small air bubbles that are present in the liquid hair color.
[0247] (See FIG. **42**A while envisioning said figure featuring the present hook **2***a*′ and squeegee **101***f*′. Do this while considering the following.) Another benefit of the present liquid control factor is a clean initial priming of the device just after a color container **51***a* has been loaded and just before device use. After a color container **51***a* has been loaded and the manifold **53***a* engaged, the operator must squeeze the handle **20***a*, thereby pushing some of the hair color out of the color container **51***a* and into the manifold **53***a*, manifold hoses **53***b* and applicator nozzle **11***a*. Since the closed position of the hook **2***a*′ and applicator nozzle **11***a* is sealed to the viscous hair color, but the air can escape, the operator may squeeze the handle **20***a* moderately, for a period of time, and all of the air from the applicator delivery system will escape through the applicator nozzle **11***a* but the color will not dispense from the closed position of the hook **2***a*′ and applicator **11***a*. This is the action of ‘priming’ the device before use.
[0248] Yet another benefit of the present liquid control factor arises in the context of an iteration of the device featuring multiple hooking applicators **1***a*. In this case, as an operator is making a pass of the device close to the hairline and only has room to place one of the two or more hooking applicators **1***a* against the scalp, the operator is free to entrain hair into one of the outside hooking applicators **1***a* and dispense color onto only that one entrained section without the mess of color dispensing from the other unoccupied hooking applicators **1***a* as the operator makes the pass.
[0249] As an operator completes a pass with the device, and releases the handle **20***a*, there will be residual color left in the applicator nozzle **11***a* and color channel **2***b* of the hook **2***a*′. As the operator prepares for a second pass of the device, the operator will place the device on the scalp of the recipient and entrain hair. If the amount of residual color left in the applicator nozzle **11***a* and color channel **2***b* is excessive, the hook **2***a*′will close over the applicator nozzle **11***a* and a portion of the excessive residual color will be squeezed out in an uncontrolled manner from between the hook **2***a*′ and applicator nozzle **11***a*. This causes an inconsistency of the color application onto the entrained hair through successive passes of the device. A solution to this problem is to create a vacuum inside of the manifold **53***a*′ in order to produce a suction that draws a portion of the residual color out of the applicator nozzle **11***a* between passes. Described below is a simple mechanical process that utilizes a manifold **53***a*′ with a flexible backing **53***j* that is compressed by the squeezing of the handle **20***a* and decompressed by releasing the handle.
[0250] A manifold **53***a*′ with a flexible backing **53***j* is achieved in the present way by replacing the (see FIG. **42**A) vertically positioned sections of the ‘L’ shaped manifold hoses **53***b* with a (see FIG.

44A) flat rectangular box that occupies approximately the same position and area as the three vertical sections of the manifold hoses **53***b* combined. Then reposition the manifold couplings along the front bottom of the rectangular box while maintaining the manifold intake in the same position as the previous manifold. ridged rear wall of the manifold **53***a* with a thin sheet of rubbery material such as silicone rubber or a similar visco-elastic material and attaching a plate to the outside of the visco-elastic sheet. Now, a mechanism that is actuated by the squeezing of the handle **20***a* will press on this manifold compression plate **53***k* creating a flexible manifold **53***a′* of reduced volume. Conversely, when the handle **20***a* is released, the compression plate **53***k* springs back outward by the tension of the flexible backing **53***j* and the volume of the flexible manifold **53***a′* increases creating suction or vacuum within the flexible manifold **53***a′*.

[0251] The following is a description of a mechanical arrangement that, when arranged appropriately within the series of mechanical operations that are actuated by the squeezing of the handle **20***a*, gives the device the ability to suction away a portion of the residual hair color left in the and applicator nozzle **11***a* after each pass of the device. (See FIGS. **44**A-D for the following.)

[0252] As the handle **20***a* is squeezed, the distal front edge of the bottom handle section **20***c* establishes contact with the manifold slide lever **119** (see FIG. **44**B). As the handle **20***a* continues to be squeezed, the manifold slide lever **119** pivoting on the latch rod **121**, establishes contact with the manifold compression plate **53***k* that is in front of and contiguous to flexible backing **53***j* of the flexible manifold **53***a′*. Continuing to squeeze the handle **20***a* causes the compression plate **53***k* of the flexible backing **53***j* to compress a small distance approximately flatly into the back of the flexible manifold **53***a′* (see FIG. **44**C). At this point in the handle **20***a* squeeze action, the compressing of the back of the flexible manifold **53***a′* causes the volume inside the flexible manifold **53***a′* to decrease. The contact rails **120** of the manifold slide lever **119** are pivotally oriented with the angle of approach of the front edge of the bottom handle section **20***c* in such a way as to allow the manifold slide lever **119** to maintain a consistent compression position of the compression plate **53***k* against the flexible backing **53***j* as the handle **20***a* is squeezed from the point of first compression contact (see FIG. **44**C) through to the point where the handle **20***a* is squeezed to the fully closed position (see FIG. **44**D). Through this compression operation, the body of the flexible manifold **53***a′* is held from the opposite side against the pressure of the manifold slide lever **119** by the manifold brace **122**. As the handle **20***a* is released, the front edge of the bottom handle section **20***c* slides down along the contact rails **120** of the manifold slide lever **119** thereby allowing the manifold slide lever **119** to back away from the compression plate **53***k*. The compression plate **53***k*, therefore, flexes back outward to the resting position causing the volume of the flexible manifold **53***a′* to increase. In summary, squeezing the handle **20***a* causes an ordered series of mechanical operations to engage. The first increment of handle **20***a* squeeze action causes the hooks **2***a′* to pivot to the closed position. The next increment causes the volume of the flexible manifold **53***a* to decrease. The last increment of handle **20***a* squeeze action causes the liquid color to dispense from the color container **51***a* and into the flexible manifold **53***a* of less volume. Releasing the handle **20***a* causes this series of mechanical operations to disengage in reverse order. As the operator releases the handle **20***a*, the pressure on the color container **51***a* ceases. The next increment of handle **20***a* release causes the compression plate **53***k* to flex back out to the resting position thereby creating the vacuum that suctions a portion of the liquid color back out of the applicator nozzle **11***a*, and therefore, the closed chamber created by the hook **2***a′* and the applicator nozzle **11***a* in the when closed position. Finally, the last increment of handle **20***a* release causes the hooks **2***a′* to return to the open position with enough color removed from the applicator nozzle **11***a* and color channel **2***b* of the hook **2***a′* that residual color left in the applicator nozzle **11***a* and color channel **2***b* is no longer an issue when the hook **2***a′*closes back over the applicator nozzle **11***a* at the beginning of the next pass.

[0253] The following is a description of a variation of the device that is suited for use on one's self, though it is not limited to this type of use (see FIG. **45**A-C for the following). This iteration employs a flat plate **123***a* in place of a rounded hook, and a tube that is flat on the bottom side as an applicator **124***a*. Said flat bottomed applicator **124***a* is closed at the distal end and features one or several openings located on said flat bottom side. These openings serve as the apertures **124***b* through which the liquid color flows. This flat bottom side of the applicator **124***a* serves as the receiving surface for the top surface of the hinged and pivotal plate **123***a*. In use, the flat plate **123***a* and flat bottomed applicator **124***a* reciprocate relative to one another between an open and closed position as does the previously described hook and applicator arrangement (FIG. **43**A-C, **1***a*). The purpose for this reciprocating action is also the same as said previously described hook and applicator arrangement. Said shared purpose is to sandwich a section of hair between the top surface of the flat plate **123***a* and the receiving surface of the flat bottom applicator **124***a*. A functional action that is not shared between the two said mechanical arrangements is, the way a section of hair **1***b* becomes sandwiched between the plate **123***a* and the flat bottom applicator **124***a* as compared to the hook and applicator arrangement **1***a*. The plate **123***a* and flat applicator **124***a* arrangement requires the operator to manually choose a section of hair **1***b* from a parting of one's own hair **1***c* and then manually place said section of hair between the plate **123***a* and flat applicator **124***a*. Once this manual selection and placement of hair into said position is achieved, the operator then squeezes the lever **125***a* of the handle/device body **126***a* and the plate **123***a* pivots on the plate hinge **123***b* into the closed position against the flat applicator **124***a* with the section of hair between **1***b*. As the operator continues to squeeze the lever **125***a* and begins to slowly pull the device away from the scalp, the mechanical means and sequence of said means that apply the color onto the sandwiched section of hair are similar in function to the previously described hooking applicator (see FIG. **43**A-C, **1***a*).

[0254] (Compare FIG. **43**A-C with FIG. **45**A-C for the following.) In addition to the shared general function and purpose of the present plate **123***a* and flat applicator **124***a* relative to the previously described hook **2***b*′ and applicator nozzle **11***a*, the two said arrangements also share the foam seal **124***c* and rubber squeegee **124***d* color flow control features and action. The difference between said foam seal **124***c* and squeegee **124***d* of the present iteration and the foam seal **11***b* and squeegee **101***f*′ of the previously described applicator nozzle **11***a* is in shape. The hooking applicator **1***a* features a rubber squeegee **101***f*′ with a rounded bottom edge as well as a foam seal **11***b* that is oriented to the rounded cylindrical shape of the applicator nozzle **11***a*. Comparatively, the top and bottom edges of the rubber squeegee **124***d* featured on the flat applicator **124***a* are likewise straight, along with a foam seal **124***c* that is oriented to the flat bottom surface of said flat applicator **124***a*.

[0255] (See FIGS. **43**A-C and **45**A-B for the following description of comparative squeegee seating.) Another difference of note between said arrangements is how the outside bottom edges of each type of squeegee **124***d*, **101***f*′ is sealed and immovable in the closed position, while each squeegee recess **124***e*, **101***h* retains the forward movement required for the squeegee action. As the plate **123***a* moves upward on the plate hinge **123***b*, the outside edges of the present squeegee **124***d*, lower into the squeegee edge seats **123***c* located on the top surface of the plate **123***a*. With the outside edges of the squeegee **124***d* secure against movement and leakage, the present squeegee recess **124***e* is likewise secure to distribute liquid hair color onto the sandwiched sections of hair consistently in like manner to the previously described hooking applicator **1***a* squeegee, in the closed position, sealing arrangement **101***f*′, **101***h*.

[0256] The following is description of the present device in use as well as a description of the mechanical actions that are engaged while using the present plate and applicator iteration of the device.

[0257] (See FIGS. **46**A and **46**B for the following) Before the operator can begin the treatment using the device, the operator must open the body **126***a* of the device by manually pressing inward on the release tabs **126***d* of the body **126***a*. One release tab **126***d* is located on either side of the device body **126***a*. Each release tab **126***d* is a longitudinal partial separation of the front of the device body rear section **126***c*. Each release tab **126***d* is partially separated from the device body

rear section **126***c* by two open longitudinal slits that extend back parallel to one another for a short distance along the front of the device body rear section **126***c*, thereby forming a slightly flexible tab **126***d* that is attached to the front of the device body rear section **126***c*. The front end of each tab **126***d* extends forward a short distance beyond the front edge of the device body rear section **126***c* and features a wedge **126***e* that is tapered at the front and extends back to a raised lip thereby forming said wedge **126***e* at the front of each tab **126***d*. Said tabs **126***d* of the device body rear section **126***c* snap accurately into the tab seat openings **126***f* located at the rear of the device body front section **126***b*. This occurs as the operator slides the two telescopically oriented device body sections **126***b*, **126***c* together. In the closed position the front body section **126***b* and the rear body **126***c* section are held in firm and accurate mating to one another by two opposing radial body closure lips **126***g*. One lip **126***g* extends around the rear, inside surface of the front body section **126***b* and the other lip **126***g* extends around the front outside surface of the rear body section **126***c*.
[0258] The operator, having released the rear body section **126***c* from the front body section **126***b*, will see a puncture fixture **127***a* protruding from the inside of the front body section **126***b*. The following is a description of the color pouch that will be pressed securely onto this puncture fixture **127***a*.
[0259] (See FIGS. **47**B-C for the following) Each color pouch that will be described features an exit port **128***d* with a puncture seal **127***c* fastened over the top of the exit port **128***d*. Said puncture seal **127***c* may be fashioned from a variety of material including foil or plastic. FIG. **47**A shows a pouch **128***a*.
[0260] (See FIG. **47**B for the following.) The operator now mixes a dual pouch **128***b* of two component bleach. The dual pouch **128***b* is preferably vacuum sealed and is large enough in size to contain the appropriate, premeasured amount of chemical components while maintaining an amount of slack within the outside pouch walls **128***e*. This slack is important for the thorough manual knead mixing required of such packaging and this while not damaging the outside pouch wall **128***e*. This dual pouch **128***b* is therefore mixed by squeezing the dual pouch **128***b* in a kneading motion until the internal pouch **128***f* with the liquid developer **128***g* bursts and the powder bleach **128***h* and liquid developer **128***g* are thoroughly mixed. This internal pouch **128***f* bursts without bursting the external pouch **128***e* because the walls and seams of the internal pouch **128***f* are assembled to be weaker and the walls tighter than the external pouch **128***e*. Also, at least one side or seam edge of the internal pouch **128***f* is joined to at least one side or seam edge of the external pouch **128***e* to avoid the now empty, internal pouch **128***f* from moving forward inside the external pouch **128***e*, while in use, and therefore, blocking the flow of liquid color from the exit port of the pouch **128***b*.
[0261] (See FIG. **47**C for the following.) The double pouch **128***c* is two separate pouches joined by a neck **128***i*. Both joined pouches **128***c* are preferably vacuum sealed with some wall slack as mentioned in the previous pouch description. An exit port **128***d* extends from the top of one of the pouches. A removable, preferably plastic, paper clip type pinch barrier **128***j* separates the two joined pouches **128***c* from prematurely mixing through the open neck **128***g* that connects the two pouches **128***c*. The bleach powder **128***h* is contained within the pouch without the exit port **128***d* and the liquid developer **128***g* is contained within the pouch that has the exit port **128***d*. Before mixing these chemical components, the operator removes the clip barrier **128***j* from the neck **128***g* joining the two pouches and empties the pouch that contains the developer **128***g* into the pouch that contains the powder bleach **128***h*. This may be accomplished by pinching the entire width of the pouch containing the developer **128***g* between two fingers and sliding the fingers from the exit port **128***d* side of the pouch to the bottom of the pouch until the entire contents of the developer pouch **128***g* is contained within the powder bleach pouch **128***h*. The barrier clip **128***j* is now placed back over the neck **128***g* and the pouch that now contains both components is kneaded as described above. Having thoroughly mixed the components, the barrier clip **128***j* is once again removed, the entire mixture is pinched back into the exit port **128***d* pouch and the barrier clip **128***j* replaced once

more. The empty pouch can simply be folded over the full pouch and placed into the device in this manner or the empty pouch may be cut off. This double pouch **128***c* design lessens the chance that the two components become prematurely mixed do to damage during shipping and handling. This design also eliminates the possibility of unmixed powder clogging the exit port **128***d* while mixing.
[0262] After the operator prepares the color pouch **128***a-c*, the operator will then push the sealed exit port **128***b* of the pouch **128***a-c* onto the puncture fixture **127***a* until the operator feels the lip inside the exit port **128***g* engage with the puncture fixture lip **127***b* at the rear of the puncture fixture **127***a*. Then the operator fits the engaged color pouch **128***a-c* into the rear body section **126***c* and closes the rear body section **126***c* over the front body section **126***b* until the release tabs **126***d* snap into tab seat **126***f* openings. The operator will now need to prime the device for use. This priming procedure is simply pressing on the lever **125***a* until color dispenses from the applicator **124***a* before beginning treatment. While the sections are shown for the embodiment of FIG. **45**A, the hooking applicator embodiment could also employ the releasably attachable sections.
[0263] First, the operator will select a functionally appropriate sized section of hair from a parting of the operator's own hair at the scalp with one hand. The parting may be drawn with the parting stem of the device and then the top portion of the parting may be preferably clipped away prior to the hair section selection. The operator will slide the fingers down the length of the selected section until the fingers are holding said section taut and by the ends. With the other hand, the operator holds the device by the body **126***a*, with the thumb of the same hand resting on the lever **125***a*. The operator then moves the device into place so that the handled section of hair is positioned between the plate **123***a* and the applicator **124***a* (while in the open position), preferably at the root of the handled section of hair. The operator then begins to press down on the lever **125***a*.
[0264] (See FIGS. **46**B-D for the following.) The first increment of distance that the lever **125***a* drops causes the plate **123***a* and applicator **124***a* to close over the section of hair. This occurs as the bottom of the lever **125***a* urges downward onto plate closure slide **129***a*. As this dual prong plate closure slide **129***a* lowers, each plate pivot end **123***d* seated within each closure slide pivot seat **129***b* (each pivot seat **129***b* is located at the bottom of each prong of the closure slide **129***a*), is also urged to lower. Consequently, as the plate pivot ends **123***d* lower, the plate **123***a* rises toward the applicator **124***a*. As the closure slide **129***a* continues to slide downward, each plate pivot end **123***d* is urged pivotally out of each closure slide pivot seat **129***b* to a position where each pivot end **123***d* is completely unseated. This unseated pivot end **123***d* position causes the top of the plate **123***a* to be in the fully closed position against the bottom of the applicator **124***a*, effectively sandwiching the section of hair between the plate **123***a* and the applicator **124***a*. Through the remaining description of the present series of mechanical motions, each pivot end **123***d* remains unseated and the plate **123***a* remains closed against the applicator **124***a* with the section of hair sandwiched between, as each plate closure slide **129***a* prong continues to slide downward against each pivot end **123***d*.
[0265] (See FIGS. **48**A-B front view for the following) The second increment of distance that the lever **125***a* lowers involves a second mechanical action by the plate closure slide **129***a*. As said slide **129***a* continues to lower, each stem compression slope **129***c* located on the inside of each prong of the plate closure slide **129***a* urges inward on each hinged flap **130***c* located on either side of the rear of the stem **130***a*. This rear section of the stem**130***a* features an open square formed into each side of the rear of the stem **130***a*. A rubber tube **130***b* with a matching outside diameter to the inside diameter of the stem **130***a* is positioned and fastened to the inside of the stem **130***a* in a manner that allows the rubber tube **130***b* to be exposed through each open square of the stem **130***b* yet the entire stem **130***b* remains sealed from leakage. This rear section of the stem **130***b* may be widened diametrically larger internally than the remaining section of the stem **130***a* to accommodate a section of rubber tube **130***b* that is larger, creating a widened chamber section of said stem **130***a*. The inside surface of each hinged flap **130***c* is positioned flush against the outside of the exposed rubber tube **130***b* section. Considering this orientation of each hinged flap **130***c*, and considering that each hinged flap **130***c* is hinged to the top of each open square of the stem **130***a*, as

each stem compression slope **129**_c_ urges inward on each hinged flap **130**_c_, the section of the rubber tube **130**_b_ that each hinged flap **130**_c_ is seated against compresses inward. This effectively creates a stem **130**_a_ of lesser volume than the stem **130**_a_ in the uncompressed state.

[0266] Now, with a section of hair held sandwiched between the plate **123**_a_ and the applicator **124**_a_ and with the stem **130**_a_ held compressed into a lower volume, the operator continues to press down on the lever **125**_a_ of the device. (See FIGS. **46**B-C for the following.) As the lever **125**_a_ lowers yet another increment of distance, the downwardly extruding distal front of the lever **125**_a_ contacts the pair of curved push arms **131**_a_. As the distal top of each push arm **131**_a_ is urged downward, the distal rear of each push arm **131**_a_ extends into the front of the body of the device **126**_a_ guided inward by each push arm slide seat **131**_b_. Adjacent to the distal rear of each push arm **131**_a_, and seated inside the front of the device body **126**_a_, is the spring action pump piston **132**_b_. Said piston **132**_b_ has front and back slightly outwardly flaring skirts, both being of annular configuration and being adapted for snug sliding contact with the interior surface of the pump chamber **132**_a_. The piston **132**_b_ also reciprocates back and forth over a centrally fixed receiving tube **132**_c_. The central, cylindrically shaped receiving tube opening of the piston is likewise adapted for snug sliding contact with the exterior surface of the receiving tube **132**_c_. This snug fit of the piston **132**_b_ against the inside walls of the piston chamber **132**_a_ prevents liquid from leaking into the space behind the piston **132**_b_ as it moves forward and pressurizes the liquid in front of the piston **132**_b_.

[0267] The receiving tube **132**_c_ is centrally attached to the inside of the front wall of the pump chamber **132**_a_. Said receiving tube **132**_c_ is open to, and serves as an extension of the tubular interior passage of the stem **130**_a_. The receiving tube **132**_c_ extends toward the rear pump chamber wall **132**_d_ a distance from the inside of the front wall of the pump chamber **132**_a_ and is distally open, said opening thereby facing the rear pump chamber wall **132**_d_ a slight distance from said wall **132**_d_. A separate short section of tube is centrally attached to the outside of the rear pump wall. This puncture fixture tube **132**_e_ is the same diameter as the receiving tube **132**_c_ and, as described previously, features a conically shaped, cage-like puncture fixture **127**_a_ at the distal end of puncture fixture tube **132**_e_. This puncture fixture tube **132**_e_ also contains a ball valve **132**_f_. Another ball valve **132**_f_ is located inside the receiving tube **132**_c_.

[0268] Now, as the operator continues to press down on the lever **125**_a_, and the distal ends of the push arms **131**_a_ press against the piston **132**_b_, the piston **132**_b_ moves forward against the tension of the piston spring **132**_g_ and creates fluid pressure inside the pump chamber **132**_a_. This fluid pressure pushes the ball of the ball valve **132**_f_ located inside of the puncture fixture tube **132**_f_ toward the rear of the device and into the closed position thereby preventing the flow of the pressurized liquid from re-entering the pouch. As the receiving stem **132**_e_ is open at the distal rear to the pump chamber **132**_a_, this same fluid pressure pushes the ball of the ball valve **132**_f_ located inside of the receiving tube **132**_c_ toward the front of the device and into the open position. Now, as the operator continues to press on the lever **125**_a_, the push arms continue to press on the piston **132**_b_ and the liquid hair color is directed only into the receiving tube **132**_c_ and, consequently, into the applicator **124**_a_. So, as the operator maintains an appropriate pressure on the lever **125**_a_, the operator may now draw the device slowly away from the operator's own head and thereby dispense the liquid hair color onto the sandwiched section of hair.

[0269] At the end of this pass of the device, the operator will release the lever **125**_a_ and the above described series of mechanical operations will disengage in reverse order. As the operator releases the lever **125**_a_, the color flow to the applicator **124**_a_ stops and the piston spring **132**_g_ pushes the piston **132**_b_ in the reverse direction thereby creating a liquid suction. This suction pulls the ball of the ball valve **132**_f_ located in the puncture fixture tube **132**_e_ toward the front of the device and therefore, to the open position. Concurrently, the ball of the ball valve **132**_f_ located inside of the receiving tube **132**_a_ is pulled toward the rear of the device and therefore, to the closed position. Now, as the piston **132**_b_ continues to move in reverse, the liquid color flows out from the color pouch into the widening space between the piston **132**_b_ and the rear pump chamber wall **132**_d_, and not back out from the receiving tube **132***c*. Once the lever **125***a* has lifted away from the push arms **132***a*, the plate closure slide **129***a* has also been lifting against the tension of the closure slide springs **129***d*. This causes the stem compression slope **129***c* (see FIGS. **48**A-B front view) portion of the plate closure slide **129***a* to lift away from the rubber tube section **130***b* of the stem **130***a* thereby creating a suction inside the stem **130***a*. This suction draws a portion of the liquid hair color back out of the applicator **124***a* just prior to the plate **123***a* lowering away from the applicator **124***a*. (See FIGS. **46**B-D for the following) Finally, the plate closure slide **129***a* lifts back up to the resting position against the tension of the closure slide springs **129***d*, causing each plate pivot end **123***d* to slide back into each closure slide pivot seat **129***d*, thereby, returning the plate **123***a* to the open resting position relative to the flat applicator **124***a*. The entire liquid hair color application process described above may be repeated by the operator until the desired number of liquid color treated sections of hair has been achieved.

[0270] Another variation of the embodiment described above (see FIG. **46**B) involves the replacement of the plate **123***a* and flat applicator **124***a* portion of the front of the stem with the previously described hook and applicator arrangement; therefore, all of the serial mechanism disclosed in the plate **123***a* and flat applicator **124***a* description above will remain the same with the exception of the previously described hooking applicator (see FIGS. **43**A-C, **1***a*) along with minor modifications of the mechanism that engages the hook **2***a′* into the closed position against the applicator nozzle **11***a*. A difference in device use may also be noticed as the operator may use the automatic hair entrainment feature novel to the hooking applicator **1***a* rather than manually selecting a section of hair (although the operator may also manually select a section of hair and place it between the hook **2***a′* and applicator nozzle **11***a* as well).

[0271] As stated, the difference in this iteration compared to the previously described plate **123***a* and flat applicator **124***a* iteration is the mechanism that engages the hook **2***a′*. (See FIGS. **49**A-C for the following) The present iteration utilizes a worm gear **133***a* positioned over a hook shaft **133***b*. A gear pin **133***c* is fixed to the shaft **133***b* and extends through the worm gear thread **133***d*. A worm gear slot **133***e* is formed into the rear of the worm gear **133***a*. A hook pivot bracket **133***f* is formed into what was above referred to as the plate pivot end **123***d* (see FIG. **46**B). Said hook pivot bracket **133***f* seats contiguous within said worm gear slot **133***e*. In use, the hook pivot bracket **133***f* is urged downward by the what is now the hook closure slide **129***a′* and the hook pivot bracket **133***e* pushes against the worm gear **133***a*. The worm gear **133***a* then slides forward over the hook shaft **133***b*. This forward slide of the worm gear **133***a* causes the worm gear thread **133***d* to push on the gear pin **133***c* thereby causing the hook shaft **133***b* to rotate. This hook shaft **133***b* rotation causes the hook **2***a′* to rotate into the closed position over the applicator nozzle **11***a*. When the lever **125***a* of the device is released, and because of the addition of the hook pivot channel **133***g* and the worm gear slot pin **133***h*, the reverse of this mechanical action causes the hook to return to the open resting position.

[0272] The next variation of the previously described plate and flat applicator iteration (see FIGS. **45**A-C) is the present hair slice color applicator (see FIGS. **50**A-E). This variation features both an elongated plate **123***a′* and flat applicator **124***a′*. The elongated plate **123***a′* features a downwardly sloped point **134***a* at the distal open end and a parting tooth receiving opening **134***b*. The elongated flat applicator **124***a′* features a parting tooth **134***c*. These features are the only differences from the previously described plate and flat applicator iteration described previously (see FIGS. **45**A-C).

[0273] The purpose of this iteration is to give the operator the option of using such a device to easily and consistently apply liquid color to longer continuous slices of hair along a parting of hair rather than intermittent variegated bundles of hair as per the variation of the device that features a plurality of hooking applicators as described previously.

[0274] (See FIGS. **50**A-E for the following) In use, this variation is preferably used by an operator on the head of hair of another person. In practice an operator draws a parting of hair **1***c* with the parting stem **27** and clips the top portion of the hair parting out of the way. The operator then places the plate point **134***a* against the bottom of the parting of hair **1***c* a functionally appropriate distance below the parting (in order not to pick up more hair than the device can thoroughly and consistently apply liquid color to in a single slow pass) and pushes said plate point **134***a* under the hair (see FIG. **50**B). The operator now pushes the plate point **134***a* across the parting of hair **1***c* maintaining a consistent distance from the line of the parting until the initial edge of the lifted hair that has slipped onto and across the plate reaches the edge of the plate support **123***e* (see FIG. **50**C for hair slice). This initial edge of the lifted hair reaching the edge of the plate support **123***e* determines the appropriate width of the hair slice on the plate support **123***e* side of the plate **123***a*′. The applicator tooth **134***c* determines the appropriate width of the slice of hair from the plate point **134***c* side of the plate **123***a*′. The applicator tooth **134***c* determines this, now, total width of the hair slice **1***i* as the plate rises to meet the applicator. As this lifting of the plate **123***a*′ occurs, the applicator tooth **134***c* penetrates through the hair slice **1***j* that has lifted onto the plate **123***a*′ and therefore separates the hair **1***j* intended for treatment by the plate **123***a*′ and applicator **124***a*′ on one side of the tooth **134***c* from the hair unintended for treatment on the other side of the tooth **1***j* (see FIG. **50**D). (See FIG. **50**A for the following) Said tooth **134***c* extends down from the applicator **124***a* from a fixed position near the inside edge of the foam seal **124***c* so that the outside edge of the hair slice **1***i* is very close to the inside edge of the foam seal **124***c* as the inside edge of the foam seal **124***c* determines the outside boundary of liquid color application onto the sandwiched slice of hair **1***i* on the tooth **134***c* side of the applicator **124***a*′. (See FIGS. **50**A and **50**D for the following) In summary, the appropriate width of each hair slice **1***i* treated is consistently determined on the plate support **123***e* side of the plate **123***a*′ by the inside edge of said plate support **123***e*, and likewise determined on the plate point **134***a* side of the plate **123***a*′ by the applicator tooth **134***c*.

[0275] (See FIG. **50**E for the following) Finally, as the operator presses on the lever **125***a* and draws the device away from the parting of hair **1***c* with a slice of hair **1***i* sandwiched between the plate **123***a*′ and applicator **124***a*′, the hair that was separated out of treatment by the applicator tooth **134***c* simply falls away from the plate **123***a* as the separated hair slides down the plate point **134***a* slope.

[0276] The final description of an iteration of the device (see FIG. **51** for the following) is an update of the comb version of the device. Referring to the earlier description of the comb iteration of the device depicted in FIGS. **30-32**, most of the mechanical elements described in these drawings remain the same in the present updated version of the comb iteration. The changes depicted here include the incorporation of the squeegee **124***d*′ as well as the changes in the geometry of the hook **103***a* aspect of this iteration that support the proper function of the squeegee **124***d*′ in applying color to the bundles of hair that fall between the applicable comb teeth **104***a*. These additional geometric elements of the hook described here are the squeegee edge seat **123***c* and the hook color channel **2***b*′. May it suffice to say that when the slide member **103***b* moves to the side thereby bringing the hook **103***a* to the closed position against the foam face **108***d* of the applicator **108***a*, with a bundle of hair between, the color application process will be very similar to the color application process described in the above plate **123***a* and flat applicator **124***a* version of the device described previously (see FIGS. **45**A-**45**B).

[0277] The final description of an iteration of the device features a liquid hair color pump arrangement that may be adapted to any of the preceding variations of the device described herein. What will be presently described is a dual pump arrangement. This arrangement mechanically functions as described previously in the first plate and flat applicator description.

[0278] The significant difference is, in instances where the both color components are liquid (which is very common in the overall hair color industry) that there are two separate pumps that may simultaneously pump each of the two components, a different component for each pump, into a stem containing a removable mixing louver rod. The benefit of this iteration is that the chemical reaction of the hair color remains freshly mixed for each pass of the device thereby achieving a more consistent process throughout the application service. Another benefit is that this color component mixing method drastically reduces waste as the components stay separated until they are mixed in the stem. The louver mixing rod is removable for easier cleaning of the device. In essence, this dual pump device would just have two hair product containers and two mechanisms to convey the hair product to the mixing louver rod and the output of the mixed hair product is then conveyed to the hooking or clamping applicator as described above.

[0279] The following describes a new use for the present invention. A hair texturizing service may be performed using any previously described hair color variegation iteration of the device. Rather than dispensing hair color with the device, the operator may load the device with any kind of a hair product, e.g., styling products. For instance, the device may be loaded with hair gel instead of hair color and said gel may be applied in a likewise variegated manner to a whole or partial head of hair. Having completed this variegated hair gel service and, furthermore allowing the gel to dry in the hair, the completed service renders a head of hair a unique variegated texture with intermittent rigid gathered sections dispersed throughout the loose untreated hair. The recipient of this texturizing service may now further arrange the hair into various styles. This variegated texturizing service may be administered using the present invention using all manner of liquid styling products, including but not limited to oils, fixatives, pomades, conditioners and protectors. Any one of said types of styling products may also contain color pigment. Permanent texturizing products, such as permanent wave solution may also be applied in the present manner.

[0280] The various hooking applicator and plate applicator variations, whether used singularly or in plurality, that are described throughout this description may be adapted to the application of a wide variety of liquids to a wide variety of head hair as well as eyelashes, Wigs, and pieces and extensions, and more generally, any elongated flexible members such as fibers, cords, wire, string, animal hair and so on, that require a process of liquid coating. The hooking and plate applicator arrangements described herein may be incorporated into various manufacturing processes to achieve a variety of one as well as two or more chemical reactive coatings that, once applied in any of the previously described manners to said variety of fibers, cords, wires, and so on, may render a coating, to said receiving structures, of various properties including visco-elastic, solid, semi-solid, insulating, conductive, etc.

Claims

1. In a device for selectively entraining hair strands from the scalp having at least one hooking applicator, the at least one hooking applicator employing a hook that rotates to entrain the hair strands, a hair product container having hair color therein and a way to apply the hair color to the entrained hair stands, the improvement comprising: a) the at least one hooking applicator is configured to form a channel between a surface of the hook and a surface of a body portion of the at least one hooking applicator having a hair product outlet therein to better accommodate high viscosity liquid hair color; b) the at least one hooking applicator include at least one foot and an opening, the at least one foot having a scalp contacting surface of continuous and flat length; the hook is positioned with respect to the at least one foot to form a scissors action between the hook and the at least one foot when the hook rotates for entraining hair strands and a portion of the hook enters the opening; c) wherein the hook and/or the at least one hooking applicator includes a recess to hold entrained hair and permit hair product to fill the recess be applied to entrained hair; d) a trigger mechanism that first rotates the hook to entrain hair strands and then squeezes the hair product container to apply hair color to the entrained hair via the at least one hooking applicator; wherein e) the at least one hooking applicator includes a flexible facing that is adjacent to the channel, and the hook has a hook offset along an edge thereof, the hook and flexible facing positioned with respect to each other so that the hook offset receives an edge of the flexible facing when the hook rotates to entrain hair strands.

2. The device of claim 1, wherein the trigger mechanism includes a mechanism for creating a suction after the squeezing of the hair product container to draw residual hair product from the at least one hooking applicator after a hair product application.

3. The device of claim 2, wherein the trigger mechanism includes a flexible manifold between the hair product container and the at least one hooking applicator, the means for creating suction comprising a plate on the flexible manifold and a pivotable lever, movement of the lever cause by movement of a handle of the trigger mechanism, pressing on the handle for introducing hair product to the at least one hooking applicator moving the lever in a first direction so as to press on the plate and reduce the volume of the flexible manifold, releasing of the handle moving the lever in an opposite direction to increase the volume of the flexible manifold and create a suction to draw hair color from the at least one hooking applicator into the flexible manifold.

4. The device of claim 1, further comprising a rear body section and a front body section, the rear body section and front body connection releasably attachable to each other, the rear body section adapted to receive the hair product container, and the front body section adapted to contain the mechanism providing the hair product to the at least one hooking applicator.

5. A method of applying a hair product to hair comprising using the device of claim 1 to apply the hair product.

6. The method of claim 5, wherein an individual with hair to be entrained for hair product application manipulates the device for hair product application.

7. The method of claim 5, wherein the hair product is a hair coloring product.

8. A method of applying a coating to one or more flexible elongated members comprising providing the device of claim 1 and using a coating material in the hair product container and applying the coating material to the one or more flexible elongated members using the device.

9. The method of claim 8, wherein the one or more flexible elongated members are fibers, cords, wire, string, and animal hair.

10. The method of claim 9, wherein the device uses a dual pouch and the dual pouch contains two chemical coatings for application to the at least one elongated flexible member.